Computing & Robotics

Year 2

Cambridge Early Years

Student Manual

PRIME BOOKS

an imprint of Prime School

Computing & Robotics

Year 2 · Cambridge Early Years · Student Manual

First published 2026 by Prime Books, the publishing imprint of Prime School, Rua Prime, Portugal.

An independent publication. This book is original Prime School material, written and illustrated for Prime School by its own teaching staff. It is designed to support the teaching of Computing and Robotics in Lower Primary. It is **not** an official Cambridge Assessment International Education publication, it is not endorsed by, affiliated with or approved by Cambridge Assessment International Education, and it is not published by Oxford University Press, Cambridge University Press or any other educational publisher.

A note on the internet addresses in this book. Every QR code and web address printed here was checked and reachable at the date of publication, and each one is printed in readable words as well as in code so the page still works with no device at all. Prime School is not responsible for the content of any site it does not itself publish. An adult should look at a page before a child does.

Safety. The activities in this book were written to be done in a classroom with a teacher present. No activity asks a pupil to enter personal information, to install anything, or to contact anybody outside the class.

TITLE	Computing & Robotics, Year 2, Student Manual	PROGRAMME	Cambridge Early Years, Lower Primary
AGES	6–7	EDITION	First edition, 2026
ISBN	978-989-33-7420-7	LANGUAGE	British English
PRINTED	on paper from responsible sources		

PRIME BOOKS · PRIME SCHOOL · PRIMESCHOOL.PT

WELCOME

Hello again, maker

Fen, Pip and Cog are waiting at the start of the path. There are five stones to cross this year.

You are in Year 2. You already know that a computer is a tool and that people give it jobs. This year you get more careful, and more exact.

Last year you learned to give instructions. This year you learn to make them **precise**. Precise means clear and exact: how many, which way, and when to stop. A robot cannot guess what you meant.

The five stones

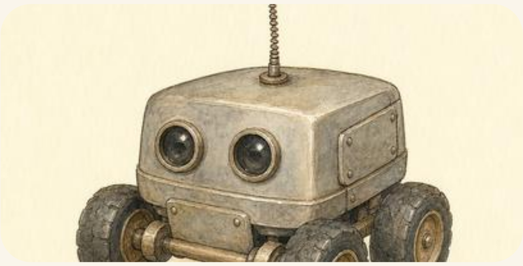
The path across the year has five stepping stones, one for each unit. The colour at the foot of each page tells you which stone you are on.

Say it, then write it

Always say your answer out loud first. Then write it. Talking first is not a warm-up, it is how the thinking gets done.

THE ONE IDEA TO REMEMBER

A computer does **exactly** what you tell it. Not what you meant.



Cog does exactly what it is told. Every single time.

That is why this book is full of things going slightly wrong. Every time, the instructions were followed perfectly, and the instructions were wrong. Finding out where is the best part.

The makers

The six of them, at the start of the year.

You will see these six on every page. They are learning exactly what you are learning, and they get things wrong first, which is the point of them.

	**Fen** a red fox	Types fast, and then has to find out why it went wrong. He is the one who learns that a robot does exactly what you say.
	**Pip** a brown hare	Thinks before she touches anything. She always says what she thinks will happen first, and she is usually close.
	**Cog** a floor robot	Not alive, and it does not think. It follows the steps it is given, exactly, every single time, even when they are wrong.
	**Wren** a hedgehog	Sorts, counts and checks. If there is data to collect, Wren has already made a table for it.
	**Moss** a badger	Mends things and keeps the basket of parts. He knows what is inside a machine because he has had one open.
	**Skylark** a barn owl	Sees the whole yard from above and carries the messages. She is the one who knows how a note gets from here to there.

Fen's russet
the colour of a red fox in summer

Fen's denim
his work jacket, soft and worn

Pip's moss green
her quilted jacket, on every page

Fen and Pip wear the same clothes in every picture in this book, on purpose, so you always know who you are looking at.

Prime School
for teachers and families
primeschool.pt

The path across the year

Five stones, one for each unit. The colour at the bottom of every page tells you which one you are on.

Before you start

Read **How to use this book** with your teacher, then **Getting set up**. After that, begin at the terracotta stone.

At the end

There are five smaller projects along the way, one big project at the end, three term checks, and a glossary of every word this book uses.

HOW TO USE THIS BOOK

What the panels mean

Every unit is built from the same pieces. Once you know these, you can find your way around any page in the book.

Skylark watches for the panels that ask you to think.

Out on the path

The start of a unit. What we are about to do, and why it matters.

What Fen tried

Fen told Cog to go to the shelf. Cog drove straight into the bin.

A short story about something going wrong, or nearly right. Every unit opens with one.

HOW TO USE THIS BOOK · CONTINUED

By the end you will

- know what this unit is teaching you
- see it before you have to do it
- find out how you will know you can do it

WAKE YOUR THINKING UP

Wake your thinking up. A quick game before the work starts. Nobody writes anything down.

Do you remember?

A question about something you already learned. If you cannot remember, look back: that is allowed and it is not cheating.

Let us find out

The teaching. Short sentences, and a picture that does the explaining. Read it with your teacher.

HOW TO USE THIS BOOK

What the panels mean

☞ **Watch me try 1**

Put these in order: soap, rinse, wet hands, dry.

1. Wet hands comes first, because soap needs water.
2. Then soap and rub.
3. Then rinse the soap off.
4. Dry last, or your hands are still wet.

We found out: the order is wet, soap, rinse, dry, and every step needs the one before it

NO SCREEN

1 One numbered thing for you to do. The number is so your teacher can say which one.

How do you know?

Every topic asks this at least once. Saying **how** you know is the difference between debugging and guessing.

☞ **Predict, then test**

Will Cog stop on the star?

I think 1	I ran it 2	I compare 3
Write or draw what you think will happen	What really happened?	Was it what you thought?

HOW TO USE THIS BOOK · CONTINUED

Find the bug

THE GOAL

Cog reaches the star.

THE PROGRAM

↑ ↑ → ■

WHAT HAPPENED

Cog stopped one square short of the star.

What is the bug? Change ONE thing, then test it again.

Skylark's own notes

A true thing that is interesting rather than needed. Skylark collects them.

No screen needed

An activity with **no device at all**: cards, chalk, paper or your own feet. There are a lot of these on purpose, because your class does not have a tablet each.

HOW TO USE THIS BOOK · CONTINUED

▣ What you need

one pencil, four paper cards, a partner

☵ Work it out together

Something you do with a partner or a group. There is always a **What you need** panel on the same page.

! Stay safe and kind

How to keep yourself, your friends and the machines safe. Never skip this one. If something feels wrong, stop and tell a trusted adult.

→ Go a little further

Something extra, if you finish early and want more.

⚑ Path challenge

The hardest thing on the page. You are not expected to get it first time.

HOW TO USE THIS BOOK · CONTINUED

ALONG THE WAY

A project

At the end of a unit there is a project: a bigger job you plan, build, test and show. There are five along the way and one big one at the end of the book.

Every page is built from these same pieces.

WORDS WE USED

command
one instruction

robot
a machine that follows steps

plan
steps before you build

mat
the squares a robot drives on

HOW TO USE THIS BOOK · CONTINUED

☐ Check what you can do

1. A short check at the end of a unit.
2. Nothing here repeats a task you have already done.

☐ say what I can do now, in my own words

	confident	almost there	needs more work
Look what I can do	☐	☐	☐

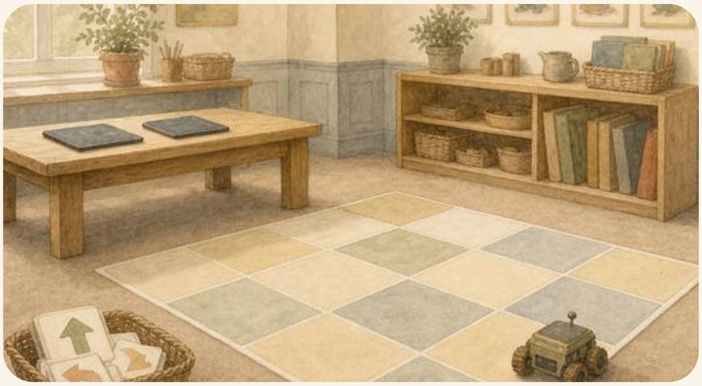
Computing with no computer at all
for teachers, when the tablets are in use
classic.csunplugged.org, activities

For teachers

Every panel on this spread appears in the units, and the build refuses to ship the book if a panel is used anywhere without being shown here. Pupils who meet an unexplained feature on page ninety have no way to find out what it wants of them.

GETTING SET UP

What is in the room



A Year 2 computing corner, ready. Notice that the tablets are face down and switched off until they are needed.

Your class will not have a tablet each, and that is completely fine. Most of the best computing in this book happens with **cards, chalk and a partner**. When it is your turn on a device, you will know what you are going to do with it before you pick it up.

▣ WORDS WE USED

tablet
draw, write, code

laptop
the bigger jobs

floor robot
follows a path

floor mat
the squares it drives on

arrow cards
plan with no screen

camera
keep a real picture

pencil
still the fastest tool

paper plan
where a program starts

The buttons you will actually use

on | **←** | **→** | **↑** | **↓** | **go** | **stop**

Most floor robots have these seven. Find them before you start, and press them gently.

Stay safe and kind

Carry a tablet flat, with two hands, and put it down before you talk to somebody. Keep food and drink away from every device in the room. Never pull a cable out by the wire: hold the plug.

No screen needed

Before any lesson with a device, your teacher may say **no screens today**. That is a real computing lesson, not a second-best one. Every idea in this book can be done on the floor with cards first, and it is usually easier to see that way.

How we work

Six rules for the whole year. Read them with your teacher, then say them in your own words.

OUR SIX RULES

1. Gentle hands: two hands on a tablet, soft taps.

→

2. Take turns: one person talking at a time.

→

3. Ask kindly: use a question, not a grab.

→

4. Help thinking, do not do a friend's work for them.

→

5. Stop and tell: if something feels wrong, tell a trusted adult.

→

6. Save your work, and leave the machine ready for the next person.

WAKE YOUR THINKING UP

Wake your thinking up. Your teacher says a rule number. You act it out with no words at all. Can the class guess which one?

Safe or not safe?

I carry the tablet flat, with two hands.	**YES** / NO
I run across the room holding a robot.	**YES** / NO
I wait for my turn on the mat.	**YES** / NO
I tell my best friend my school password.	**YES** / NO
I put my drink down on the table beside the keyboard.	**YES** / NO
I ask before I open something new.	**YES** / NO

1 Write two of the six rules in your own words. NO SCREEN

2 Say who your trusted adult is at school. NO SCREEN

My trusted adult at school is

▣ What you need

this page, a pencil, and a partner to talk to

! Stay safe and kind

Rule five is the most important one in this book. If anything on a screen worries you, upsets you or asks you a question about yourself, you do not have to deal with it on your own. Stop, and tell a trusted adult straight away. You will never be in trouble for telling.

ScratchJr, and how to install it

for teachers, before Unit 2

scratchjr.org

For teachers

The rules above are written for pupils to restate rather than recite. Asking a class to reword rule four, about helping a friend think rather than doing the work for them, is worth ten minutes early in the year and it settles most of the paired work that follows.

Computational thinking and programming

The Terracotta Stone

4 topics: What an algorithm is, Using an algorithm, What an error is, Finding an error

A robot does exactly what you say, not what you meant.

UNIT 1 STARTS HERE

Clear steps, and what happens without them

Out on the path

This unit is about giving instructions that actually work. You will write them, test them, find the one that is wrong, and fix it. That loop is what computing is.

What Fen tried

*Fen wanted Cog to drive to the bookshelf. He said, **go over there**, and pointed. Cog drove straight forward, past the shelf, and stopped against the bin.*

Cog did nothing wrong. It cannot see Fen pointing, and it does not know what **there** means. It only had one instruction: forward.

By the end you will

- say what an algorithm is, in your own words
- write steps that are clear and exact
- say what will happen before you test it
- find the one step that is wrong, and change only that one

Four cards, one job. Pip is putting the last one in place.

THE LOOP THIS WHOLE UNIT USES

1 Plan it → 2 Say what will happen → 3 Test it → 4 Find the bug → 5 Change one thing

More algorithm and debugging lessons
for teachers, to plan the next lesson
barefootcomputing.org

1.1 · WHAT AN ALGORITHM IS

An algorithm is clear steps for a job

WAKE YOUR THINKING UP

Wake your thinking up. Your teacher says a job: put on a coat. The class shouts the steps in order. Anybody may shout **stop** if a step is missing.

Let us find out

An **algorithm** is a clear set of steps for a job. You already use them. Washing your hands is an algorithm. Lining up is an algorithm.

The important word is **clear**. A step is clear when there is only one way to do it.

AN ALGORITHM YOU ALREADY KNOW

1. Wet your hands →
2. Soap and rub →
3. Rinse →
4. Dry

Clear, or not clear?

Not clear	Clear	Why
Move a bit.	Move forward two steps.	How far is a bit?
Turn.	Turn right.	Which way?
Go over there.	Walk to the blue shelf and stop.	Where is there?
Get it.	Pick up the red book.	Get what?

☞ **Watch me try 1**

Fen says: get the book somehow. Make it clear.

1. Say where to go: walk to the blue shelf.
2. Say what to pick up: pick up the blue book.
3. Say where to go next: walk back to your seat.
4. Say when to stop: stop.

We found out: four clear steps replaced one unclear one, and now only one thing can happen

✷ **Skylark's own notes**

The word algorithm is about twelve hundred years old. It comes from the name of a mathematician, al-Khwarizmi, who wrote books of clear steps for working things out. He had never seen a computer, because there were not any.

How do you know?

Pip says **move a bit** is a clear step, because she knows what it means. Is Pip right? How do you know?

1.1 · YOUR TURN

Write clear steps

1 Which one is clear? Draw a ring around it. NO SCREEN

Move a bit	Move forward two steps

2 Which one is clear? Draw a ring around it. NO SCREEN

Turn	Turn left

3 Number these steps for putting on a coat, from first to last. NO SCREEN

1	2	3	4
close the zip	put one arm in	pick up the coat	put the other arm in

4 Make this clear: **tidy up somehow**. NO SCREEN

My clear step is

5 Write four clear steps for packing your school bag. NO SCREEN

6 Write four clear steps for sharpening a pencil safely. NO SCREEN

7 Say your steps for task 6 to a partner. Your partner does exactly what you say, and nothing else. Did it work?

NO SCREEN

It worked first time.	**YES** / NO
My partner had to guess something.	**YES** / NO
I found a step I had missed.	**YES** / NO

8 Which step in task 6 was the hardest to make clear? Why?

NO SCREEN

What you need

this page, a pencil, a partner

No screen needed

Everything on this page is done with a pencil and a partner. There is no device in topic 1.1 at all, because an algorithm is not a computer thing: it is a thinking thing, and a computer is only one place you can use one.

1.1 · YOUR TURN

Whose steps are clearer?

9 Two algorithms for the same job: walk to the door. Which is clearer?

NO SCREEN

FEN WROTE

1. Go to the door
2. Stop

PIP WROTE

1. Stand up
2. Push your chair in
3. Walk to the door
4. Stand behind the last person

10 Say why. Use the word **clear**. NO SCREEN

Pip's is clearer because

11 Fen's algorithm is shorter. Is shorter always better? NO SCREEN

12 Write an algorithm for a job in your classroom. Give it to a partner. Do not tell them what the job is. NO SCREEN

13 Your partner follows your steps. Could they tell what the job was?

NO SCREEN

What you need

paper, a pencil, and four people

Work it out together

In fours. One person writes an algorithm for making a jam sandwich. The other three follow it **exactly**, with no guessing at all. Stop the moment a step could mean two things.

Path challenge

Write an algorithm that a person can follow but a robot could not. What is in it that a robot cannot do?

→ Go a little further

Count the steps in your algorithm. Now write the same job in fewer steps, without making any of them unclear. What did you join together?

For teachers

Pupils will offer 'be careful' and 'do it properly' as steps. Both are worth accepting and then testing: ask the class to act out 'be careful' and let them discover that nobody can, because it names a manner rather than an action. That is the distinction topic 1.1 is really teaching.

1.2 · USING AN ALGORITHM

Say what will happen, then test it

Do you remember?

What is an algorithm? Say it in your own words before you read on.

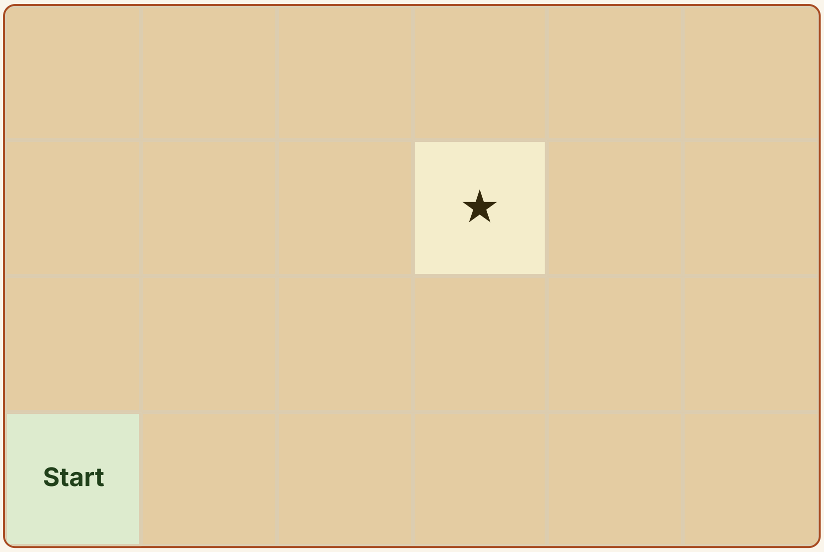
Let us find out

Before you run an algorithm, say what you think will happen. That is a **prediction**.

Saying the steps out loud without running them is a **dry run**. A dry run finds most mistakes before you have moved anything at all.

Fen dry-runs his plan at the edge of the mat. Cog has not moved yet.

COG STARTS HERE, FACING UP THE MAT



Cog always starts facing up the page. One command moves it one square.

FEN'S COMMANDS

↑ ↑ → → → ■

☞ Predict, then test

Where will Cog stop? Mark it on the mat with a pencil cross.

I think 1
Write or draw what you think will happen

I ran it 2
What really happened?

I compare 3
Was it what you thought? Did it match your cross?

☞ **Watch me try 2**

Dry-run Fen's commands with your finger.

1. Up one: Cog is on square 1 across, 2 up.
2. Up again: 1 across, 3 up. Cog is level with the star.
3. Right three times: 2, then 3, then 4 across.
4. Stop. Cog is on 4 across, 3 up.

We found out: the commands do reach the star, and the dry run proved it before anything moved

How do you know?

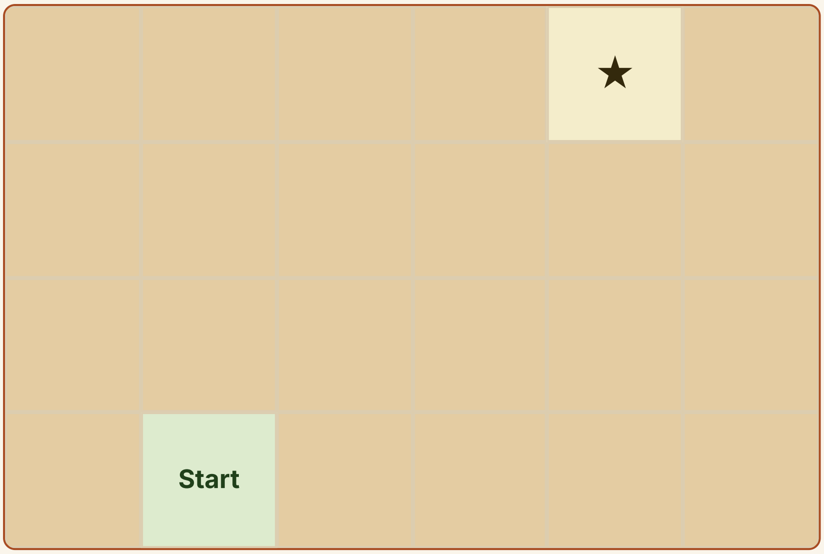
How do you know Cog will not drive off the mat? Point at the square that tells you.

1.2 · YOUR TURN

Plan a path

NO SCREEN

1. Draw Cog's path on this mat with a pencil.



2 Write the commands for your path. Use arrows. NO SCREEN

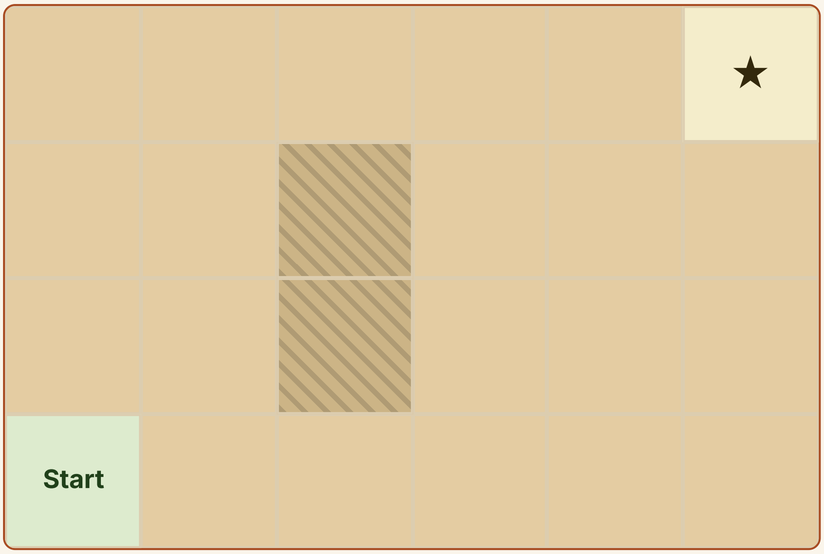
YOUR COMMANDS

3 Say what will happen. Then follow it with your finger. NO SCREEN

4 Did your finger land on the star? NO SCREEN

My finger landed on the star.	**YES** / NO
I needed one more command.	**YES** / NO
I had one command too many.	**YES** / NO

5 Now a bag is on the mat. Plan a path around it. NO SCREEN



The shaded squares are the bag. Cog cannot drive through it.

6 Write the commands for the path around the bag. NO SCREEN

YOUR COMMANDS

7 Count your commands. Write the number. NO SCREEN

I used

8 Can you reach the star in fewer commands? Try again. NO SCREEN

▣ **What you need**

a pencil, a rubber, this page

Stay safe and kind

If you are using a real floor robot, keep your fingers away from the wheels while it is moving, and walk beside it rather than running. Clear the bags and coats off the floor before you start.

1.2 · YOUR TURN

Patterns in a path

Let us find out

Sometimes a path repeats. The part that repeats is called the **unit**. If you can see the unit, you can say what comes next without checking every square.

THE UNIT HERE IS UP, RIGHT ↑ → | ↑ → | ↑ →

The line shows where the unit starts again.

9 What comes next? Draw it in the empty box. NO SCREEN

↑ ↑ → | ↑ ↑ → | ↑ [ ] [ ]

10 Write the unit that repeats. NO SCREEN

The unit is

11 Make your own repeating path. Draw the unit twice. NO SCREEN

12 Swap with a partner. Can they say your unit? NO SCREEN

13 Why do programmers look for patterns? One sentence. NO SCREEN

Skylark's own notes

A floor robot has no idea where it is. It does not know it is on a mat, or in a classroom, or in Portugal. It only knows the last command it was given. Everything else is in your head, which is why the plan matters more than the robot does.

→ Go a little further

A path repeats up, right, up, right for eight commands. Without drawing it, how many squares up will Cog end up? Say how you worked it out.

Path challenge

Plan a path that ends exactly where it started. What has to be true about the turns?

1.3 · WHAT AN ERROR IS

A bug is a mistake in the steps

Cog has stopped with one wheel over the line. Pip is working out why.

Let us find out

A **bug** is a mistake in the steps. **Debugging** means finding it and fixing it.

A bug is not a mistake in you. Every programmer in the world writes bugs, all day, and then finds them. That is the job.

WHY A BUG IS ALWAYS IN THE STEPS

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT
your steps	→	the robot follows them exactly	→	something happens

Three kinds of bug

What went wrong	The bug	Example
A step is missing.	missing step	The square has only three sides.
The steps are in the wrong order.	wrong order	You dry your hands before you rinse them.
A step says the wrong thing.	wrong step	You said left when you meant right.

☞ Watch me try 3

Cog should stop on the mat. It drove one square too far.

1. What was the goal? Stop on the mat.
2. What did we write? Forward, forward, forward, stop.
3. What happened? Cog went one square past.
4. Which kind of bug? One forward too many, so a wrong step.

We found out: there was one extra forward, so the fix is to remove one command and test again

How do you know?

Fen says the robot made a mistake by itself. Is Fen right? How do you know?

! Stay safe and kind

Nobody in this class laughs at a bug. Finding one is the good part, and a person who says they never write bugs has not written much.

1.3 · YOUR TURN

Name the bug

missing step | wrong order | wrong step

1. Goal: wash your hands. Steps: wet, soap, dry, rinse. Name the bug. NO SCREEN

2. Goal: draw a square. Steps: line, line, line. Name the bug. NO SCREEN

3. Goal: turn to the window on your right. Step: turn left. Name the bug. NO SCREEN

4. Goal: clap twice. It clapped three times. Name the bug. NO SCREEN

5. Match each bug to its fix. NO SCREEN

missing step	swap two steps over
wrong order	add the step that is not there
wrong step	change what one step says

6 Write a set of steps for a job. Now put ONE bug in it on purpose.

NO SCREEN

7 Give it to a partner. Can they find your bug?

NO SCREEN

8 Can they say which kind of bug it is?

NO SCREEN

My partner found the bug.	YES / NO
My partner named the kind of bug.	YES / NO
My partner found a bug I did not mean to put in.	YES / NO

9 Fen says a bug means you are bad at computing. Is Fen right?

NO SCREEN

10 Pip says the computer made the mistake by itself. Is Pip right? Say what really happened.

NO SCREEN

11 Name this bug. The steps say turn left. The door is on the right.

NO SCREEN

12 A friend has just found a bug in their own work. Write one kind sentence to say to them.

NO SCREEN

What you need

paper, a pencil, a partner

Work it out together

In pairs. Each of you writes four steps for getting a book off the shelf, with one bug in it. Swap. Find the other's bug without being told anything. Then say which kind it was.

Skylark's own notes

The very first computer bug that got written down was, for once, a real one. In 1947 a moth got into a machine called the Harvard Mark II and stopped it working. They taped the moth into the logbook and wrote **first actual case of bug being found**. The moth is still in the book, in a museum.

1.4 · FINDING AN ERROR

The five questions

Fen and Pip go through the plan one line at a time.

Let us find out

When something goes wrong, do not change things at random. Ask these five questions, in this order, every time.

THE DEBUG CHECKLIST

1. What was the goal?

→

2. Are the steps in the right order?

→

3. Is every word clear?

→

4. Did I miss a step?

→

5. Change ONE thing, then test again.

WHY YOU ONLY EVER CHANGE ONE THING

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT	WHAT IS KEPT
change one thing	→	test it	→	did the goal happen?	if not, ask the five questions again

☞ Watch me try 4

Cog should reach the star. It stopped one square short.

1. The goal: stop on the star.
2. The order: up, up, right, right. That looks right.
3. The words: all clear, all arrows.
4. A missing step? The star is three across, and there are only two right arrows.
5. Change one thing: add one right arrow. Test again.

We found out: one command was missing, and adding exactly one fixed it

How do you know?

Why is it a bad idea to change four things and then test? What could go wrong?

1.4 · YOUR TURN

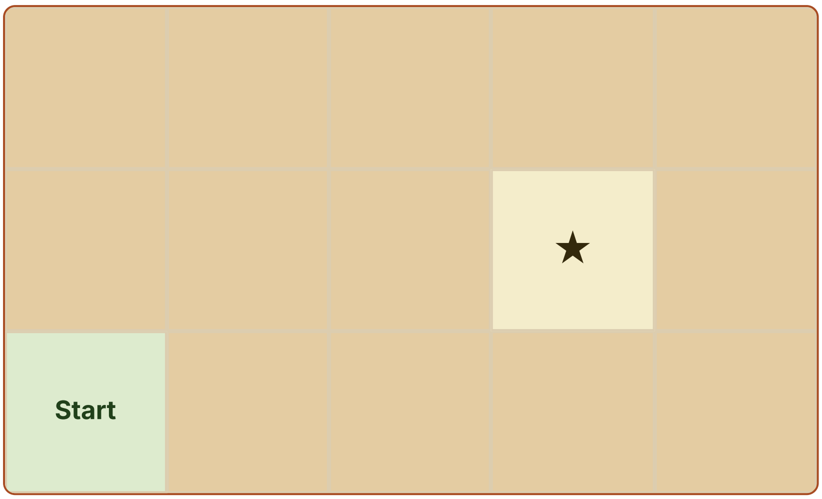
Find it and fix it

Find the bug

THE GOAL

Cog reaches the star.

THE PROGRAM



↑ → → ■

WHAT HAPPENED

Cog stopped one square short of the star.

What is the bug? Change ONE thing, then test it again.

1 Write the command you would add. NO SCREEN

I would add

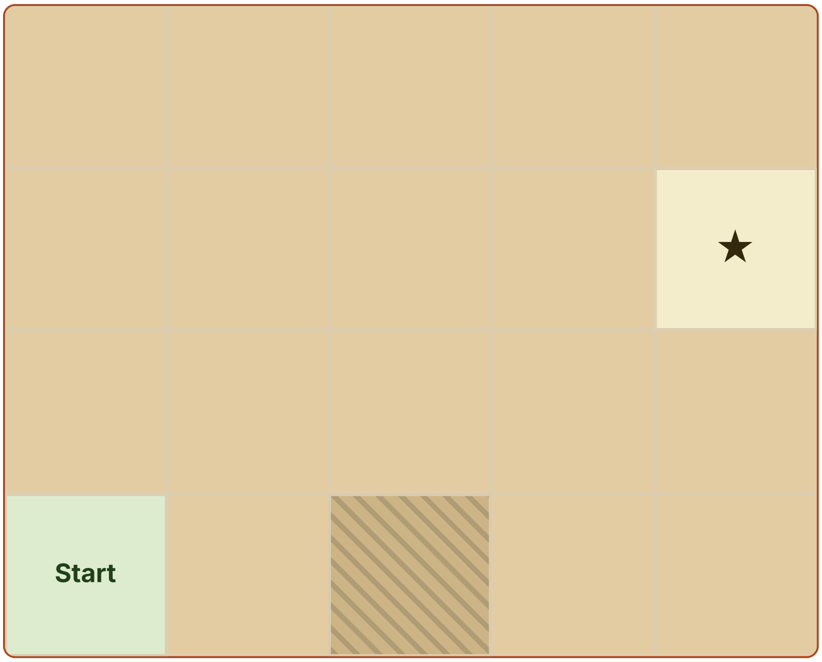
2 Where in the list would you add it? Say why. NO SCREEN

Find the bug

THE GOAL

Cog reaches the star without touching the brick.

THE PROGRAM



→ → → → ↑ ↑ ■

WHAT HAPPENED

Cog stopped against the brick and would not move.

What is the bug? Change ONE thing, then test it again.

3 What is the bug? Change ONE thing. NO SCREEN

4 Draw your fixed path on the mat above. NO SCREEN

5 Test your fix with your finger. Did it reach the star? NO SCREEN

My fix worked first time.	YES / NO
I had to change something else as well.	YES / NO

6 Break your own path on purpose. Ask a partner to fix it. NO SCREEN

▣ What you need

a pencil, a rubber, a partner

☞ Predict, then test

Before you fix bug 2, say which arrow you are going to change.

I think 1
Write or draw what you think will happen

I ran it 2
What really happened?

I compare 3
Was it the one you changed in the end?

1.4 · YOUR TURN

Say why it happened

Let us find out

A good debugger can say the **cause**: why the bug was there in the first place. That is what stops you writing the same bug again tomorrow.

The bug	The fix	The cause
Cog turned the wrong way.	change left to right	I mixed up left and right.
Cog went one square too far.	remove one forward	I counted the squares from the wrong one.
Nothing happened at all.	press go	I forgot the last thing you have to do.

7 Fill in the cause for this one: the square had only three sides. NO SCREEN

The cause was

8 Write about a bug you found today. Say the bug, the fix and the cause. NO SCREEN

9 Write your own debug checklist in three lines. NO SCREEN

10 Teach your checklist to somebody in another group. NO SCREEN

11 Look back at your prediction on the last page. Was it what really happened? Compare the two and say what was different. NO SCREEN

12 Which of the five questions found your bug fastest today? NO SCREEN

Question number

Path challenge

Fen fixed a bug and the program still did not work. He says there must be a second bug. Is that possible? How would you find out?

→ Go a little further

Keep a bug book this week. Every time you find one, write the bug, the fix and the cause. By Friday, is there a kind you keep making?

For teachers

The cause column is the part pupils skip, and it is the part that transfers. A pupil who can say 'I counted from the wrong square' has learned something reusable; a pupil who only says 'I took one arrow out' has not. Expect to model it several times before it takes.

UNIT 1 · WORDS AND CHECKS

Words we used

WORDS WE USED

algorithm
clear steps for a job

command
one instruction

robot
a machine that follows steps

mat
the squares a robot drives on

dry run
saying the steps before you test

goal
what success looks like

obstacle
something in the way

bug
a mistake in the steps

☐ Check what you can do

1. Say what an algorithm is, without using the word steps.
2. Give a partner three clear commands to reach the door.
3. Find the bug: the goal is four sides, the steps draw three.
4. Name the three kinds of bug.
5. Say the five debug questions in order.
6. Say why you change only one thing at a time.

More algorithm and debugging lessons
for teachers, to plan the next lesson
barefootcomputing.org

UNIT 1 · HOW DID YOU DO?

Look what I can do

- [ ] I can say what an algorithm is in my own words.
- [ ] I can write steps that are clear and exact.
- [ ] I can say what will happen before I test it.
- [ ] I can find a bug and name what kind it is.
- [ ] I can change one thing and test it again.
- [ ] I can say why a bug happened.

	confident	almost there	needs more work
writing clear steps	☐	☐	☐
saying what will happen first	☐	☐	☐
finding a bug on my own	☐	☐	☐
changing only one thing	☐	☐	☐
saying why the bug happened	☐	☐	☐

Cog carries a note to the shelf. That is the project at the end of this unit.

✷ Skylark's own notes

Nobody writes a program that works first time, and nobody expects to. The people who are good at this are the people who are quickest at finding out what is wrong, not the people who never get it wrong.

UNIT 2 OF 5

Coding and programming

The Ochre Stone

12 topics · Scripts, Motion, Looks, Sound, Events, Control, Sensing, Variables, Backdrops, Lists, Logic, The interface

A program is blocks joined in order, and you can always run it again.

UNIT 2 STARTS HERE

Blocks, joined in order

Out on the path

In Unit 1 you wrote steps on paper. Now you build them out of blocks, run them, watch what happens and change one thing. This is the longest unit in the book, and it is the one where you make things move.

What Pip tried

Pip put four blocks on the stage and pressed the flag. Nothing happened at all.

Her blocks were perfect. They were just not **touching**. A block that is not joined to the one above it is not part of the program, and the computer never even reads it.

By the end you will

- join blocks into a script that runs
- start a program with an event
- use a loop instead of writing the same thing twice
- keep something in a variable and in a list
- find the part of the screen you need

Three blocks joined, and one on its own. Only three of them will run.

A SCRIPT THAT RUNS

Notice how the tabs interlock. That is what joined means.

This is the tool. The thinking is still yours.

> **Stay safe and kind**
> Ask before you open an app, and never install anything yourself. If a screen asks you for your name or your age, stop and tell a trusted adult.

2.1 · SCRIPTS

A script is blocks in a line

WAKE YOUR THINKING UP

Wake your thinking up. Stand in a line of four. Hold hands. Now one person lets go. Which part of the line can still be led?

Let us find out

A **script** is blocks snapped together in a line. The computer reads them from the top, one at a time, in order.

Two rules follow from that, and they explain nearly every problem you will have this unit. Blocks that do not touch do not run. And the order of the line is the order of the job.

READ IT FROM THE TOP

green flag
move right 2
say hi

☞ Watch me try 1

Pip has three blocks that will not run. What is wrong?

1. Is there a start block? Yes, the flag is there.
2. Are the blocks in a sensible order? Yes.
3. Are they joined? No. There is a gap under the flag.
4. Snap them together, so the flag touches the move block.

We found out: the blocks were right and the joining was wrong, so nothing ran

The stage and the blocks

THE STAGE

a park with a tree on the right

THE CHARACTERS

the cat

the tree

THE BLOCKS

⚑ → ☼ ♪ ↻ ▤

YOUR SCRIPT GOES HERE

⚑ green flag

→ move right 3

Four places to know: the stage, the characters, the blocks and the script area.

How do you know?

How do you know the computer reads a script from the top and not from the bottom?

2.1 · YOUR TURN

Join them up

1 Which script will run? Draw a ring around it. NO SCREEN

green flag
move right 2

move right 2

2 Say why the other one will not. NO SCREEN

3 Number these blocks from first to last. NO SCREEN

1	2	3	4
say hello	move right	green flag	stop

4 Cut four paper tabs. Build a script on your desk. NO SCREEN

5 Swap desks. Read your partner's script out loud, from the top. NO SCREEN

6 Was there a gap in it? What would happen? NO SCREEN

Every block was joined.	YES / NO
I found a gap.	YES / NO
I could read it from the top without guessing.	YES / NO

7 Write a three-block script for a cat that moves and speaks. NO SCREEN

MY SCRIPT

8 Where does the start block go? NO SCREEN

It goes

9 Draw a ring around the block the computer reads first. NO SCREEN

10 Fen puts his say block above the flag. What happens? NO SCREEN

11 Build the same job with the blocks in the wrong order. Read it. NO SCREEN

12 Why does order matter more than how many blocks you have? NO SCREEN

▣ What you need

scissors, card for four tabs, a pencil, a partner

☐ No screen needed

All of topic 2.1 is paper tabs on a desk. A pupil who can build and read a paper script will build a working one on a screen in half the time.

2.2 · MOTION

The number is part of the instruction

Pip counts the stones out loud as Cog crosses them.

Let us find out

A move block carries a **number**. Four means four. It is not a suggestion and the computer will not round it for you.

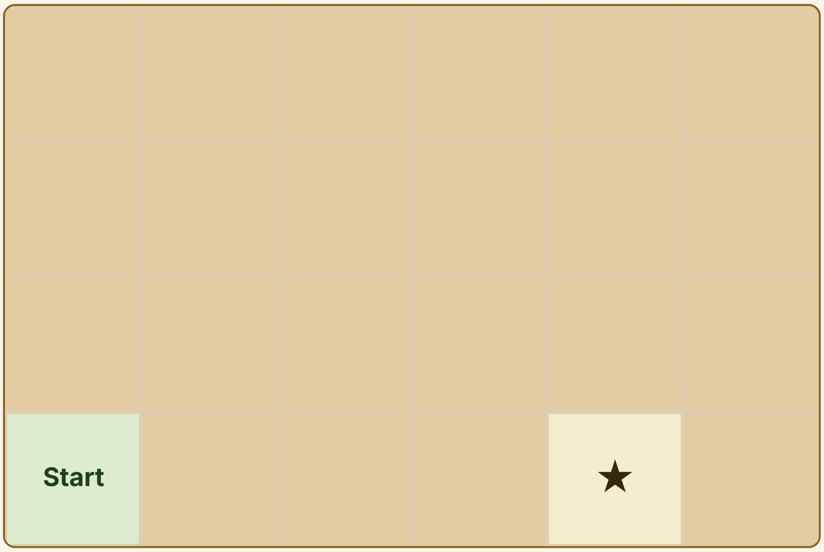
A turn block carries a direction. Left and right are different instructions, not two ways of saying turn.

THESE TWO BLOCKS ARE NOT THE SAME

→ move right 1
→ move right 4

One square, then four squares. Five squares altogether.

GETTING FROM START TO THE STAR



☞ Predict, then test

How many squares must the move block say?

I think 1

Write or draw what you think will happen

I ran it 2

What really happened?

I compare 3

Was it what you thought? Did it match when you counted?

☞ Watch me try 2

Pip wants the cat four squares right. She sets move right 5.

1. Count the squares between the cat and the tree. There are four.
2. Her block says five.
3. So the cat will go one square too far and pass the tree.
4. Change the number from five to four. Nothing else.

We found out: the block was the right block and the number was wrong, which is the commonest bug in this whole unit

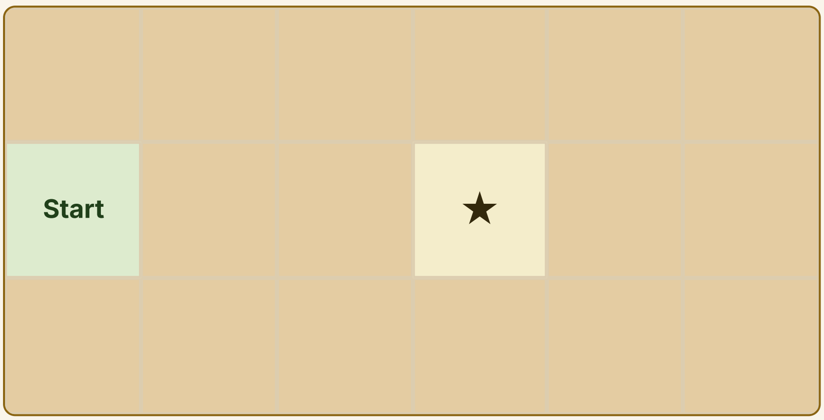
How do you know?

How do you know whether to change the block or change its number?

2.2 · YOUR TURN

Count the squares

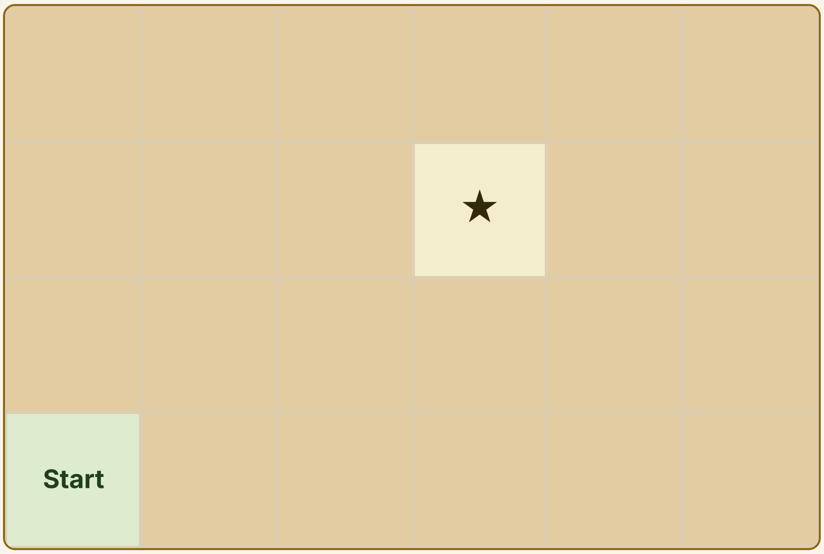
1 Write the number this move block needs. NO SCREEN



FILL IN THE NUMBER

→ move right

2 Now write the blocks for this one. NO SCREEN

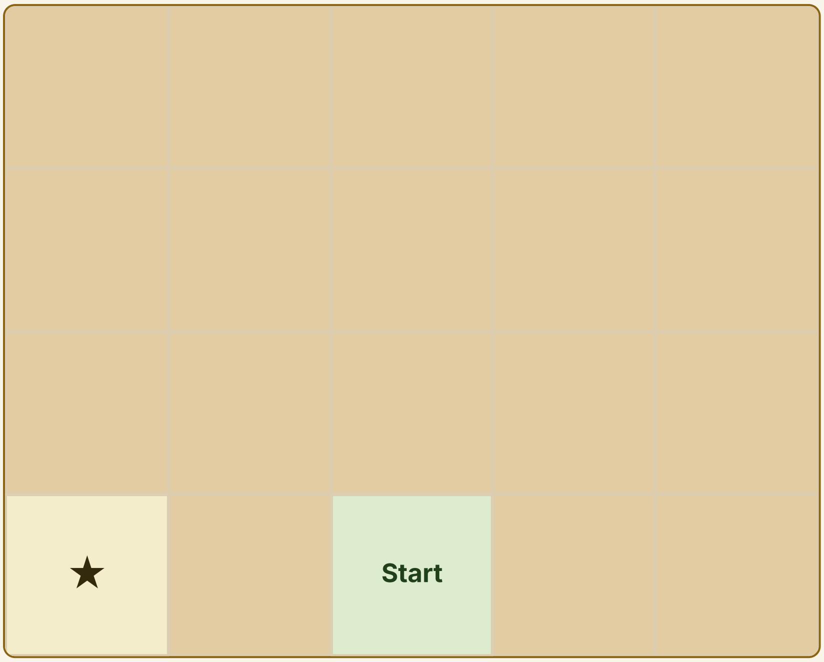


3 Say what will happen before you check it. NO SCREEN

4 Check it with your finger. Did it match? NO SCREEN

My finger landed on the star.	**YES** / NO
My number was one too big.	**YES** / NO
My number was one too small.	**YES** / NO

5 Turn left or turn right? Write the answer. NO SCREEN



6 Which two blocks are the same? Ring them. NO SCREEN

move right 3 | move right 3 | move left 3

7 Change one number so this reaches the star. NO SCREEN

→ **move right 2**
→ **move up 5**

8 A cat is told move right 0. What happens? NO SCREEN

9 Write two different scripts that end in the same square. NO SCREEN

10 Which of your two is shorter? NO SCREEN

11 Why might a shorter script be better? NO SCREEN

12 Draw a path on paper. Give the numbers to a partner to follow. NO SCREEN

What you need

a pencil, a rubber, a partner

Stay safe and kind

If you use a real floor robot, keep fingers away from the wheels and walk beside it. Clear the floor first.

2.3 · LOOKS

Changing how something looks

The same character, small and then large. It has not moved at all.

Let us find out

A **looks** block changes how a character appears. It can say something, hide, show, grow or shrink.

Here is the useful part: a looks block does not move anything. A character can change completely and still be in exactly the same place.

NOTHING MOVES IN THIS SCRIPT

- green flag
- grow bigger
- say I am here

The block	What changes	Does it move?
say hello	a speech bubble appears	no
grow bigger	the character gets larger	no
hide	you cannot see it	no
move right 3	the position	yes

☞ Watch me try 3

Fen wants the cat to speak at the tree. It speaks at the start.

1. Where is the say block? Directly under the flag.
2. Where is the move block? Under the say block.
3. So it speaks first, then travels. The order is wrong.
4. Swap the two blocks over, so it moves and then speaks.

We found out: the blocks were both correct and their order was wrong, so the cat spoke in the wrong place

How do you know?

How do you know a say block has run, if the character has not moved?

2.3 · YOUR TURN

Say it, show it, hide it

1 Does this block move the character? Ring your answer. NO SCREEN

say hello: moves | say hello: does not move

2 Sort these into the two trays. NO SCREEN

say hello | move right 2 | grow bigger | hide | move up 1

changes how it looks	changes where it is

3 Write a script where the cat speaks at the tree, not at the start. NO SCREEN

4 Say what will happen. Then read it to a partner.

NO SCREEN

5 Draw the cat before and after a grow block.

NO SCREEN

before	after

6 True or not true?

NO SCREEN

A hidden character is still there.	**TRUE** / NOT TRUE
A say block moves the character.	**TRUE** / NOT TRUE
Grow makes a character larger.	**TRUE** / NOT TRUE

7 Write what the cat should say when it arrives.

NO SCREEN

8 Put your say block in the right place in the script. NO SCREEN

9 Which looks block would make a character vanish? NO SCREEN

10 Wren says a hide block deletes the character. Is Wren right? NO SCREEN

11 Design a character that grows when it reaches the tree. NO SCREEN

12 Why is it useful that looks and motion are separate? NO SCREEN

What you need

paper, a pencil, a partner

2.4 · SOUND

Sound is an output you can hear

The sound comes out of the tablet. You can hear it, and you cannot see it.

Let us find out

A **sound** block is an output, exactly like a picture on the screen. The difference is that you hear it instead of seeing it.

Choose a sound that fits the moment. A pop when something lands is useful. A pop every half second is only noise.

HOW A SOUND GETS OUT

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT
you tap the sound block	→	the machine reads the sound file	→	the speaker moves the air

LAND, THEN POP

```
⚑ green flag
→ move right 3
♪ play a pop
```

☞ Watch me try 4

Wren puts the pop before the move. What does she hear?

1. The flag runs the script from the top.
2. The pop is first, so she hears it straight away.
3. Then the character moves, in silence.
4. She wanted the pop on arrival, so the pop must go last.

We found out: the sound played at the wrong moment because it was in the wrong place in the line, not because it was the wrong sound

How do you know?

How do you know a sound block has run if the room is noisy?

Stay safe and kind

Keep the volume low, and never put headphones on somebody else's ears without asking. If a sound is too loud, tell a trusted adult rather than turning it up to check.

2.4 · YOUR TURN

The right sound, at the right moment

1 Is a sound an input or an output? NO SCREEN

input | output

2 Number these blocks from first to last. NO SCREEN

1	2	3
play a pop	green flag	move right 3

3 Write a script that speaks and then makes a sound. NO SCREEN

4 Say what you will hear, in order. NO SCREEN

5 Match each moment to a good sound. NO SCREEN

the cat lands	a creak
the door opens	a cheer
the game is won	a pop

6 Clap a three-beat pattern. Write it with marks. NO SCREEN

MY PATTERN

7 A sound plays every half second. Is that useful? NO SCREEN

8 Name an output you can see and one you can hear. NO SCREEN

9 Draw where the sound comes out of a tablet. NO SCREEN

10 Could a program work with no sound at all? NO SCREEN

11 Why might a quiet program be better in a classroom? NO SCREEN

12 Add one sound to your script. Say why you chose it. NO SCREEN

What you need

this page, a pencil, and your own hands to clap with

Skylark's own notes

A speaker makes sound by pushing air backwards and forwards very fast. If you rest a paw gently on one while it plays, you can feel the air being pushed. Sound is something moving, not something magic.

2.5 · EVENTS

Nothing runs until something starts it

Fen reaches for the flag. Cog has not moved, and will not until he touches it.

Do you remember?

What are the two rules about a script? Say them before you read on.

Let us find out

An **event** is something that starts a script. The green flag is an event. Tapping a character is an event.

The event block always sits at the **top**. Blocks under it wait, quietly, possibly forever, until the event happens.

EVERY PROGRAM, WITHOUT EXCEPTION

1 something happens: the event → 2 the blocks under it run, in order

The event	What starts	When
the green flag is clicked	every flag script	at once
this character is tapped	only that character's script	when a paw touches it
nothing	nothing at all	never

☞ Watch me try 5

Pip's script has no event block. What happens when she runs it?

1. She presses the flag.
2. Her script has no flag block at the top.
3. So nothing tells it to start.
4. Nothing happens, and the blocks are all perfectly correct.

We found out: a script with no event never runs, however good the rest of it is

How do you know?

How do you know an event has happened, if the character has not moved yet?

2.5 · YOUR TURN

What starts it?

1 Match each event to what it starts. NO SCREEN

the school bell	that character runs
the green flag	the class lines up
tapping a character	the script runs

2 Which block goes at the top? Ring it. NO SCREEN

move right 2 | green flag | say hello

3 Write an event and three blocks under it. NO SCREEN

MY SCRIPT

4 Why must the event come first? NO SCREEN

5 Name three events at home. NO SCREEN

6 Fen says the flag makes the cat move. Is that exactly right? NO SCREEN

7 Act it out. One person is the event, three are the blocks. NO SCREEN

8 What happened before the event person moved? NO SCREEN

9 Draw a script with the event in the wrong place. NO SCREEN

10 Say what that script would do.

NO SCREEN

11 Can one program have two different events?

NO SCREEN

12 True or not true?

NO SCREEN

An event goes at the top.	TRUE / NOT TRUE
A script with no event still runs.	TRUE / NOT TRUE
Tapping a character can be an event.	TRUE / NOT TRUE

What you need

space to stand in a line of four

Work it out together

In fours. One of you is the event and stands at the front. The other three are blocks, in a line behind. Nobody may move until the event person has moved. Then each block acts in turn.

2.6 · CONTROL

A loop does the same thing on purpose

Cog goes round twice. Pip counts on two paws.

Let us find out

When the same steps happen again and again, that is a **repeat**. A repeat you planned is a **loop**.

To build a loop, first find the **unit**: the chunk that starts over. Then count how many times it happens.

THE UNIT HERE IS STEP, CLAP step clap | step clap | step clap

The lines show where the unit begins again. Three units.

SIX BLOCKS

```
→ move right 1
♪ clap
→ move right 1
♪ clap
→ move right 1
♪ clap
```

```
↻ repeat 3 times
    → move right 1
    ♪ clap
```

The same job, in a loop.

☞ **Watch me try 6**

Which of those two scripts is better, and why?

1. Do they do the same thing? Yes, exactly the same.
2. Count the blocks: six on the left, two inside the loop.
3. If you wanted ten claps, the left one needs twenty blocks.
4. The loop only needs its number changed from three to ten.

We found out: a loop does the same job with fewer blocks, and it is far easier to change afterwards

How do you know?

How do you know where a unit ends and the next one begins?

2.6 · YOUR TURN

Find the unit, then loop it

1 Write the unit that repeats. NO SCREEN

The unit is

hop | hop | stop ‖ hop | hop | stop

2 How many times does it repeat? NO SCREEN

3 Draw the next two boxes. NO SCREEN

● | □ | □ ‖ ● | □ | □ ‖ ● | |

4 How many claps does this make? NO SCREEN

↻ repeat 4 times
- ♪ clap
- ♪ clap

5 Write this out without a loop. How many blocks? NO SCREEN

6 Now write it with a loop. Fill in the number. NO SCREEN

↻ repeat times
- → move right 1

7 Make a three-part unit with your body. Do it twice. NO SCREEN

8 Write your unit down. NO SCREEN

9 A loop repeats 5 times with 2 blocks inside. How many actions? NO SCREEN

10 Change one number to double the claps. NO SCREEN

11 Why do programmers look for patterns? NO SCREEN

12 Find a repeating pattern in your classroom. Draw its unit. NO SCREEN

What you need

space to move, a pencil

Path challenge

A loop repeats a loop inside it, three times each. How many actions altogether? Draw it before you answer.

2.7 · SENSING

A program can feel the world

Cog has touched the brick and stopped. It felt it.

Let us find out

Sensing is when a program can tell that something has happened: it has touched the edge, it has touched another character, or somebody has tapped it.

Sensing is an **input**. It is the world getting into the program, rather than the program putting something out.

SENSING IS AN INPUT

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT
the cat touches the wall	→	the sensing block notices	→	the cat turns round

FEEL, THEN ACT

green flag
when touching the edge
turn right

Predict, then test

Cog drives towards a brick two squares away. What will it do?

I think 1	I ran it 2	I compare 3
Write or draw what you think will happen	What really happened?	Was it what you thought? Did it match what happened?

Watch me try 7

Cog is told to drive forward for ever, with a brick in the way.

1. Cog reads its instruction: keep going forward.
2. It reaches the brick.
3. With no sensing block, it does not know the brick is there.
4. It pushes against it and keeps trying, because that is what it was told.

We found out: without a sensing block a robot cannot notice anything at all, and it will do exactly what it was told for ever

How do you know?

How do you know a robot has sensed something, rather than just stopped?

2.7 · YOUR TURN

Notice, then act

1 Input or output? Ring your answer. NO SCREEN

touching the edge: input | touching the edge: output

2 Sort these into the two trays. NO SCREEN

say hello | when tapped | play a pop | touching the wall

the program notices	the program does something

3 Write a script that turns when it touches the edge. NO SCREEN

4 Play blindfold robot. Your partner guides you by voice only. NO SCREEN

5 Which of your senses were you using? NO SCREEN

6 Cog has no sensor. Draw what happens at the brick. NO SCREEN

7 Name one thing a floor robot cannot feel. NO SCREEN

8 Name three things a tablet can sense. NO SCREEN

9 Pip says Cog knows the brick is there. Is Pip right? NO SCREEN

10 Draw a path where sensing would help. NO SCREEN

11 Write the sensing block you would add. NO SCREEN

12 Why is sensing more useful than counting squares? NO SCREEN

What you need

a blindfold or a scarf, a clear space, a partner

Stay safe and kind

In the blindfold game the guide walks beside their partner and never lets them near a table, a chair or a step. Walk only. Stop the moment your partner says stop.

2.8 · VARIABLES

A box that holds one thing at a time

One acorn goes in the first box. There is only room for one.

● Let us find out

A **variable** is a labelled box that holds one thing at a time.

Here is the part that catches everybody. When you put a new number in, the old one is **gone**. Not underneath it, not beside it. Gone.

score

1	2	3	4
0	1	2	3

The same box, four times. Each new number replaces the last.

☞ Watch me try 8

The score box holds 4. Pip adds 1. What is in it now?

1. The box holds 4.
2. Adding 1 makes 5.
3. The 5 goes into the box.
4. The 4 is gone. There is no way to get it back.

We found out: the box holds 5 and only 5, because a variable holds one thing at a time

COUNTING TAPS

```
green flag
set score to 0
when tapped
add 1 to score
```

How do you know?

How do you know what is in a variable, if you cannot see inside the box?

2.8 · YOUR TURN

What is in the box?

1 Fill in the box after each step. NO SCREEN

score

1	2	3	4
0			

Start at 0. Add 1. Add 1. Add 1.

2 The box held 3. You set it to 7. What is in it? NO SCREEN

3 Where is the 3 now? NO SCREEN

4 Use a real box and counters. Put in 2, then set it to 5. NO SCREEN

5 What did you have to do with the 2? NO SCREEN

6 Write a name for a box that counts goals. NO SCREEN

7 Why is a good name useful? NO SCREEN

8 The box holds 5. You add 1 three times. What is in it? NO SCREEN

score

1	2	3	4
5			

9 Set score to 0 is missing. What might go wrong? NO SCREEN

10 Draw two boxes with different labels. NO SCREEN

11 Can one box hold two numbers at once? NO SCREEN

12 Why do we call it a variable and not a number? NO SCREEN

▣ What you need

a small box or cup, ten counters, a pencil

☐ No screen needed

Use a real cup and real counters for this whole topic. A pupil who has physically had to take the old counter out before putting the new one in never forgets what a variable does.

2.9 · BACKDROPS

The backdrop is the place

Two backdrops, one character. Pip decides where the story happens.

Let us find out

The **backdrop** is the picture behind everything. It is the place your story happens in.

Change the backdrop and the same character is somewhere new. That is how a story moves from one scene to the next without the character going anywhere.

HOW A SCENE CHANGES

1. scene 1: the cat in the park →
2. change the backdrop →
3. scene 2: the same cat, by the sea

The stage and the blocks

THE STAGE

a park, with a tree

THE CHARACTERS

the cat

THE BLOCKS

⚑ → ☼ ♪ ↻ ▤

YOUR SCRIPT GOES HERE

The stage holds the backdrop. The characters stand on it.

☞ Watch me try 9

Wren changes the backdrop. Has the cat moved?

1. The cat was in the middle of the stage.
2. The backdrop behind it changed.
3. The cat is still in the middle of the stage.
4. It has not moved at all, but the story has.

We found out: changing a backdrop moves the story and not the character, which is why it is the easiest way to make a second scene

How do you know?

How do you know a backdrop has changed and the character has not moved?

2.9 · YOUR TURN

Two places, one story

1. Draw two backdrops for a story. NO SCREEN

scene 1	scene 2

2. Write one sentence for each scene. NO SCREEN

3. Which comes first? Number them. NO SCREEN

4. Does your character move between scenes? NO SCREEN

5. Ring what a backdrop changes. NO SCREEN

the place	the character	the blocks

6 Cut out a paper character. Hold it against two drawings. NO SCREEN

7 Did the paper character change? NO SCREEN

8 Write the blocks that change the scene. NO SCREEN

9 Name a backdrop that would suit a scary moment. NO SCREEN

10 Name one that would suit a happy ending. NO SCREEN

11 Could two characters be on one backdrop? NO SCREEN

12 Why is a backdrop useful in a story with no words? NO SCREEN

What you need

paper, coloured pencils, scissors, and a paper character

2.10 · LISTS

Boxes in order, and position matters

Four boxes in a row, one thing in each. The fifth is empty.

Let us find out

A **list** is a row of boxes kept in order. Each box has a position: one, two, three.

Position one is not the same box as position two. That sounds obvious, and it is the whole difficulty: if you ask for position three you get position three, even when you meant the last one.

my colours

1	2	3	4	5
red	blue	green		

Three things in a five-box list. Positions four and five are empty, and empty is not the same as nothing.

☞ **Watch me try 10**

Pip asks for position 2 of red, blue, green. What does she get?

1. Count the boxes from the left.
2. Position one holds red.
3. Position two holds blue.
4. So she gets blue, whether or not she meant blue.

We found out: a list gives you the position you asked for, exactly, which is why counting from the correct end matters

How do you know?

How do you know which box is position one? Point at it.

2.10 · YOUR TURN

Count the positions

1 What is in position 3? NO SCREEN

animals

1	2	3	4
cat	dog	hen	cow

2 What position is the cow in? NO SCREEN

3 Fill in this list with four of your own things. NO SCREEN

my list

1	2	3	4

4 Ask a partner for position 2. Did they say the right one? NO SCREEN

5 Lay four objects in a row. Number the positions. NO SCREEN

6 Swap positions 1 and 4. Write the new list. NO SCREEN

7 Add one thing to the end. What position is it in? NO SCREEN

8 A list has 5 boxes and 3 things. How many are empty? NO SCREEN

9 Number this list from first to last. NO SCREEN

1	2	3	4
hen	cow	cat	dog

10 Fill in what each box holds after the swap. NO SCREEN

after the swap

1	2	3	4

Four boxes, in order. Position one is on the left.

11 How is a list different from a variable? NO SCREEN

12 Why must a list stay in order? NO SCREEN

13 Write a list of five things, with positions. NO SCREEN

What you need

four small objects, paper, a pencil

Skylark's own notes

Programmers argue about whether a list should start at position one or position zero, and both are used in real languages. Whichever you use, you have to be consistent, or you fetch the wrong box for ever.

2.11 · LOGIC

If this, then that

A fork in the path. Cog has stopped, and Fen has to decide.

Let us find out

A **condition** is a question with only two answers: yes or no.

An **if** block asks the condition, and the answer chooses the path. If the answer is no, the blocks inside are skipped completely.

```
▾ if [touching the edge] then
    → turn right
```

If it is touching the edge, turn. If not, carry straight on.

Is it touching the edge?

YES	NO
turn right	keep going

Two answers, two paths, and no third option.

☞ **Watch me try 11**

The condition is no. What do the blocks inside do?

1. The if block asks its question.
2. The answer is no.
3. So the blocks inside are skipped.
4. The program carries on with whatever comes after the if block.

We found out: when a condition is no, the blocks inside do not run at all, and the program does not stop

How do you know?

How do you know whether an if block ran, if nothing seemed to change?

2.11 · YOUR TURN

Yes or no

1 Which of these has only two answers? Ring them. NO SCREEN

Is it raining?

What is your favourite colour?

Are you touching the wall?

2 Write a condition about your classroom. NO SCREEN

3 Fill in the two paths. NO SCREEN

Is the ball red?

YES	NO

4 The condition is yes. Do the inside blocks run? NO SCREEN

5 The condition is no. Do they run? NO SCREEN

6 Play the if game your teacher shows you. Watch who moves. NO SCREEN

7 Write an if block for a cat that hides when tapped. NO SCREEN

8 Say what happens when nobody taps it. NO SCREEN

9 Number these from first to last. NO SCREEN

1	2	3
do the inside blocks	ask the question	get the answer

10 Write a condition that is always yes. NO SCREEN

11 What would happen with a loop around it? NO SCREEN

12 Why is a yes or no question easier for a computer? NO SCREEN

What you need

space to stand and sit, a pencil

Path challenge

Write an if block inside a loop, so a cat turns every time it touches the edge. Draw it before you build it.

2.12 · THE INTERFACE

Knowing where things live

The four places on the screen, seen from above.

Let us find out

The **interface** is the parts of a program you look at and touch. There are four places worth knowing, and knowing them is half of using the app.

The four places

THE STAGE

the backdrop you chose

THE CHARACTERS

the cat

the tree

THE BLOCKS

⚑ → ☼ ♪ ↻ ▤

YOUR SCRIPT GOES HERE

⚑ green flag

→ move right 2

The place	What lives there
the stage	your backdrop and your characters
the character list	every character in your program
the block palette	all the blocks you can choose from
the script area	the blocks you have joined together

☞ **Watch me try 12**

Fen cannot find his move block. Where should he look?

1. Is it on the stage? No, the stage shows the story.
2. Is it in the character list? No, that shows characters.
3. Is it in his script area? He has not dragged it there yet.
4. So it is still in the block palette, where all blocks start.

We found out: every block starts in the palette and only appears in the script area once you drag it there

How do you know?

How do you know whether a block is in your script or only in the palette?

Stay safe and kind

Everything in this app is yours and stays on your school machine. Never type your name, your age or where you live into any program, and if a screen asks you for them, stop and tell a trusted adult.

2.12 · YOUR TURN

Find your way around

1 Match each thing to where it lives. NO SCREEN

your backdrop ●	● the character list
every block you could use ●	● the stage
the blocks you joined ●	● the block palette
your characters ●	● the script area

2 Where do all blocks start? NO SCREEN

3 Draw the four places on this screen. NO SCREEN

the screen

4 Label each one. NO SCREEN

5 Where would you look for the green flag block? NO SCREEN

6 Where do you watch your program happen? NO SCREEN

7 Point at all four places on a screen or on your drawing. NO SCREEN

8 Which place would be empty in a brand new program? NO SCREEN

9 Name the buttons you would use most. NO SCREEN

The ones that matter

flag | stop | → | ↻

10 Why is it useful to know the names of the four places? NO SCREEN

11 Teach the four places to somebody in another group. NO SCREEN

12 Which place do you use most? Say why. NO SCREEN

What you need

paper, a pencil, and a screen if one is free

Step-by-step block projects to try next

for teachers and families

projects.raspberrypi.org, Scratch intro

UNIT 2 · WORDS AND CHECKS

Words we used

WORDS WE USED

script
blocks joined in order

event
something that starts a program

sequence
steps in order

variable
a box that holds one thing

list
boxes kept in order

sound
an output you can hear

condition
a question with a yes or no answer

stage
where you watch it happen

☐ Check what you can do

1. Say the two rules about a script.
2. Say where an event block goes, and why.
3. Find the unit in a pattern, then say how many times it repeats.
4. Say what happens to the old number in a variable.
5. Say what position 1 of a list means.
6. Name the four places on the screen.

Stay safe and kind
Nothing you build in this unit ever needs your real name. Save your work with the name your teacher gives you.

UNIT 2 · HOW DID YOU DO?

Look what I can do

- [ ] I can join blocks into a script that runs.
- [ ] I can start a program with an event.
- [ ] I can use a loop instead of writing the same thing twice.
- [ ] I can keep a number in a variable.
- [ ] I can find a thing by its position in a list.
- [ ] I can find my way around the screen.

	confident	almost there	needs more work
joining blocks so they run	☐	☐	☐
putting the event at the top	☐	☐	☐
finding the unit and looping it	☐	☐	☐
using a variable	☐	☐	☐
finding my way around the screen	☐	☐	☐

The two-scene story is the project for this unit.

✷ Skylark's own notes

The people who build the games you play use every single block you have met in this unit: events, loops, variables, lists and conditions. There are more of them, and they are bigger, but they are the same ideas.

✎ For teachers

Unit 2 is twelve topics and will take a term. The two that reward the most time are 2.6 loops and 2.8 variables: pupils who can find a unit and who understand that a box holds one thing at a time will meet no real surprise in Year 3. Both are best taught with counters and clapping before any screen is involved.

UNIT 3 OF 5

Managing data

The Sage Stone

5 topics · What data is, Collecting data, Sorting and grouping, Displaying data, Interpreting data

Data is what we collected, and a table lets it speak.

UNIT 3 STARTS HERE

Information you collected on purpose

Out on the path

This unit is about finding things out by counting them. You will ask a real question, collect every answer, sort what you get, draw it, and then say something true about it.

What Wren tried

Wren asked her class which fruit they liked best. Then she counted up and got twenty-eight answers, in a class of twenty-four.

Four people had answered twice. Her counting was perfect and her data was wrong, which is worse, because the chart still looked convincing.

By the end you will

- say what data is, and where it comes from
- ask a clear question and tally every answer once
- sort things by a rule anybody could follow
- draw a chart where one square means one
- read a chart and say one true thing about it

Wren has her objects and a blank chart. Nothing is counted yet.

THE FIVE STEPS OF THIS WHOLE UNIT

1. Ask a clear question →
2. Tally every answer once →
3. Total each row →
4. Draw the chart →
5. Say something true

Stay safe and kind

Ask questions about things, never about people. Favourite fruit is fine. Somebody's home, their family or how they look is not ours to collect, and we do not write it down.

3.1 · WHAT DATA IS

Data is collected, not found

WAKE YOUR THINKING UP

Wake your thinking up. Everybody stand. Sit down if you have a pet. Look round the room. You have just collected data without writing anything.

Let us find out

Data is information we collected on purpose. That last part matters: data does not lie about waiting to be found, somebody goes and gets it.

Two rules make data trustworthy. Everybody answers the **same** question. And one person gives **one** answer.

Not data yet	Data	Why
I think most like cats	18 of 24 chose cats	one is a guess, one was counted
lots of people walk to school	9 walk, 15 come by car	lots is not a number
everybody likes football	11 chose football	everybody is almost never true

☞ Watch me try 1

Wren counted 28 answers in a class of 24. What went wrong?

1. How many people are there? Twenty-four.
2. How many answers? Twenty-eight.
3. That is four more answers than there are people.
4. So four people answered twice, and the rule is one person, one answer.

We found out: the count was right and the data was wrong, because the total told Wren something the chart could not

How do you know?

How do you know a total is wrong, without asking anybody again?

3.1 · YOUR TURN

Is it data?

1 Data or a guess? Ring your answer. NO SCREEN

I think most people walk | 9 people walk

2 Sort these into the two trays. NO SCREEN

6 chose apples | lots like pears | everybody likes cake | 11 have a pet

data	a guess

3 Write a question you could collect data about. NO SCREEN

4 Write one you could not. Say why. NO SCREEN

5 Count how many people in your group have brown eyes. NO SCREEN

6 Is that data? Say why. NO SCREEN

7 A class of 20 gave 23 answers. What happened? NO SCREEN

8 Why does one person give one answer? NO SCREEN

9 Would this question work? What is your favourite thing?

NO SCREEN

10 Rewrite it so it would work.

NO SCREEN

11 Name something in your classroom you could count right now.

NO SCREEN

12 Count it. Write the number.

NO SCREEN

I counted

What you need

this page, a pencil, and your own group to ask

No screen needed

The whole of Unit 3 is paper, pencils and a class to ask. There is not one activity in it that needs a screen, which is why it is the unit to teach when the tablets are booked.

3.2 · COLLECTING DATA

One mark for each answer

Wren makes one mark as Pip answers. She does it while she listens, not afterwards.

Let us find out

A **tally** is one mark for each thing you count. Make the mark while you ask, not from memory afterwards.

Marks are grouped in **fives**: four upright strokes, and the fifth laid across them. Grouping means you can read a big count without counting every single mark.

HOW THE MARKS ARE REALLY WRITTEN

The answer	Tally	How many
apple	☍ / /	7
banana	/ / / /	4
pear	☍ ☍ / /	12

☞ Watch me try 2

Why lay the fifth mark across the other four?

1. Count these marks one at a time: it is slow and easy to lose your place.
2. Now count them in fives: five, ten, eleven.
3. The crossed mark makes each group of five visible at a glance.
4. So a tally of thirty is read in six looks instead of thirty.

We found out: grouping in fives is what makes a long tally readable, and it is why everybody writes them the same way

☞ Predict, then test

Ask four friends one question. How many marks will you have?

I think 1
Write or draw what you think will happen

I ran it 2
What really happened?

I compare 3
Was it what you thought? Did it match your total?

How do you know?

How do you know you tallied everybody exactly once?

3.2 · YOUR TURN

Collect it yourself

1 Write the number these marks show.

NO SCREEN

What we found	Tally	How many
red	☍ / / /	
blue	/ / /	

2 Draw the tally for 6.

NO SCREEN

3 Draw the tally for 11.

NO SCREEN

4 Write your question at the top of this table. NO SCREEN

MY QUESTION

What we found	Tally	How many

5 Ask everybody in your group. Tally as you go. NO SCREEN

6 Total each row. NO SCREEN

7 Add your totals. Does it match your group size? NO SCREEN

My total matched.	**YES** / NO
My total was too big.	**YES** / NO
My total was too small.	**YES** / NO

8 If it did not match, say what you would check first. NO SCREEN

9 Why tally while you ask, rather than afterwards? NO SCREEN

10 Ask the same question to a different group. NO SCREEN

11 Did you get the same answer? Should you have? NO SCREEN

12 Which was the hardest part of collecting? Say why. NO SCREEN

What you need

paper, a pencil, a clipboard if you have one, and a group to ask

Work it out together

In threes. One asks, one tallies, one counts the people. At the end, the counter's number and the tallier's total must match. If they do not, do not fix the numbers: find out why.

3.3 · SORTING AND GROUPING

A rule anybody could follow

Two trays, one rule. Wren can say exactly why each thing went where it did.

Let us find out

To **sort** is to put things into groups using a rule.

Here is the test of a good rule: two people sorting the same objects with your rule get the **same** groups. If they do not, the rule was not clear enough.

A weak rule	A clear rule	Why
the nice ones	the red ones	nice is different for everybody
big things	longer than my hand	big compared with what?
things I like	has wheels	nobody else can use your rule

ONE RULE, TWO TRAYS

has wheels	no wheels

☞ Watch me try 3

Wren sorts by nice ones. Pip re-sorts the same objects. What happens?

1. Wren puts the shell and the feather in the nice tray.
2. Pip does not think the shell is nice, and puts it in the other tray.
3. Both of them followed the rule they were given.
4. So the rule was the problem, not either of them.

We found out: a rule that two people apply differently is not a sorting rule, however reasonable it sounds

How do you know?

How do you know your rule is clear? What would you get a partner to do?

3.3 · YOUR TURN

Sort it two ways

1 Ring the rule that is clear enough to use.

NO SCREEN

the good ones | has four legs | the interesting ones

2 Write a clear rule for sorting your pencil case.

NO SCREEN

3 Sort eight objects into these two trays.

NO SCREEN

4 Write your rule above the trays. NO SCREEN

5 Ask a partner to sort the same eight with your rule. NO SCREEN

6 Did they get the same groups? NO SCREEN

We got the same groups.	**YES** / NO
We disagreed about one thing.	**YES** / NO
We disagreed about several.	**YES** / NO

7 Now sort the same eight a different way. NO SCREEN

8 Can the same objects be sorted two ways? NO SCREEN

9 Fill in this two-question sort. NO SCREEN

TWO RULES AT ONCE

	round	not round
red		
not red		

10 Which box holds things that are both? NO SCREEN

11 Which box holds things that are neither? NO SCREEN

12 Why is a Carroll diagram harder than two trays? NO SCREEN

What you need

eight small objects, two trays or two sheets of paper, a partner

Path challenge

Write a rule that puts exactly half your objects in each tray. How many rules did you try before one worked?

3.4 · DISPLAYING DATA

One square means one

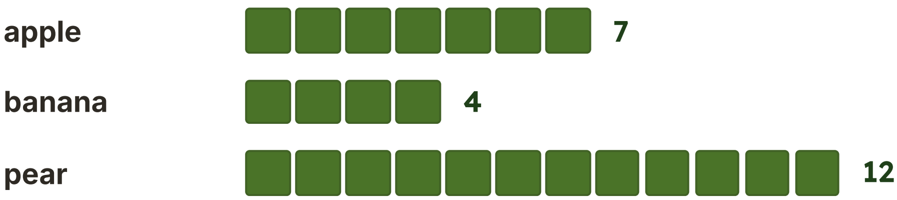
Wren and Moss look at the chart. Neither of them has to ask what it means.

Let us find out

A **chart** is a picture of your data. It is for somebody who was not there when you collected it.

There is one rule, and everything else follows from it: **one square means one**. Draw six squares for six, never five big ones.

A BLOCK CHART OF WREN'S FRUIT



one square means one

THE SAME IDEA, WITH PICTURES

cats	● ● ● ● ●
dogs	● ● ●
fish	● ●

one picture means one

☞ Watch me try 4

Fen draws a taller square for banana because he likes bananas.

1. His chart shows banana as the tallest bar.
2. But four people chose banana and twelve chose pear.
3. Somebody reading his chart would believe banana won.
4. One square must always mean one, or the chart tells a lie.

We found out: a chart that breaks the one-square rule is not a chart at all, it is just a drawing

How do you know?

How do you know somebody else can read your chart? What do they need on it?

3.4 · YOUR TURN

Draw it so somebody else can read it

1 How many does this bar show?

NO SCREEN

COUNT THE SQUARES

red  6

one square means one

2 Draw a bar for 9.

NO SCREEN

3 Draw the chart for your own data from topic 3.2.

NO SCREEN

4 Write what one square means on your chart. NO SCREEN

5 Which bar is tallest? NO SCREEN

6 Show your chart to somebody who was not there. NO SCREEN

7 Could they read it without you explaining? NO SCREEN

They read it straight away.	YES / NO
They asked me one question.	YES / NO
I had to explain it.	YES / NO

8 What would you add to make it clearer? NO SCREEN

9 Draw the same data as a picture chart. NO SCREEN

10 Which of your two charts is easier to read? Say why.

NO SCREEN

11 A chart has no key. What is the problem?

NO SCREEN

12 Why must every square be the same size?

NO SCREEN

What you need

squared paper if you have it, a pencil, coloured pencils

Sorting, counting and charts to watch

for teachers, to play on the class screen

bbc.co.uk/bitesize, primary computing

3.5 · INTERPRETING DATA

Say one true thing

Skylark taps the tallest column. She is about to say something true about it.

Do you remember?

What does one square mean on a chart? Say it before you read on.

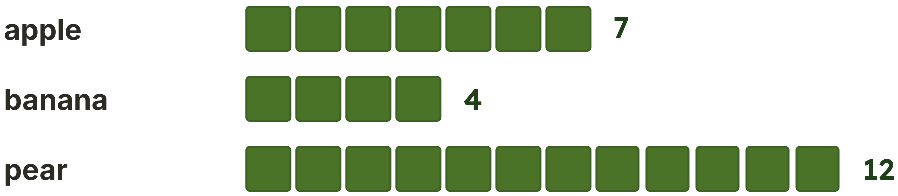
Let us find out

Collecting data is only half the job. The other half is **reading it**: saying what it shows, and being able to point at the part that shows it.

The three words you need are **most**, **least** and **the same**.

What we can say	How the chart shows it
Pear got the most	its bar is the longest
Banana got the least	its bar is the shortest
Two got the same	the two bars are equal
23 people answered	add all the bars together

READ THIS ONE



one square means one

☞ Watch me try 5

Somebody says this chart proves pears are the best fruit.

1. Does the chart show pear with the longest bar? Yes.
2. Does it show what twenty-three people in one class chose? Yes.
3. Does it show anything about anybody else? No.
4. So it shows what this class chose, not what is best.

We found out: a chart shows what was collected and nothing more, and stretching it further is where data starts to mislead

How do you know?

Somebody says most people chose apple. How would you check?

3.5 · YOUR TURN

What does it show?

HOW WE COME TO SCHOOL

What we found	Tally	How many
walk	⚯ / / / /	
car	⚯ ⚯ /	
bus	/ / / /	

1 Total each row. NO SCREEN

2 Which has the most? NO SCREEN

The most is

3 Which has the least? NO SCREEN

The least is

4 How many people answered altogether? NO SCREEN

5 Write one true sentence about this table. NO SCREEN

6 Write one sentence that is NOT true, and say why. NO SCREEN

7 Somebody says nobody walks. Point at what shows that is wrong.

NO SCREEN

8 Draw this data as a chart.

NO SCREEN

9 Two more people join and both walk. Which bar changes?

NO SCREEN

10 Does the answer to task 2 change? Say why.

NO SCREEN

11 Read your own chart from topic 3.4. Say two true things.

NO SCREEN

12 Why is it useful to show data to somebody who was not there?

NO SCREEN

▣ What you need

this page, a pencil, and your own chart from topic 3.4

→ Go a little further

Collect the same data from another class. Is the answer the same? Say what that tells you about your first chart.

! Stay safe and kind

Never write somebody's name next to their answer. A tally is about how many, not about who, and that is what keeps it fair.

UNIT 3 · WORDS AND CHECKS

Words we used

▣ WORDS WE USED

data
information we collected

tally
one mark for each thing

sort
put into groups by a rule

table
rows and columns

chart
a picture of data

most
the biggest group

least
the smallest group

total
how many altogether

☐ Check what you can do

1. Say what makes something data rather than a guess.
2. Draw the tally for thirteen.
3. Say the test of a good sorting rule.
4. Say what one square on a chart means.
5. Say the three words for reading a chart.
6. Say why a total that is too big matters.

✷ Skylark's own notes

Everything a computer holds is data: your work, a photograph, a song. It is all counted and stored the same way underneath, which is why a machine can sort ten thousand things as easily as you sorted eight.

UNIT 3 · HOW DID YOU DO?

Look what I can do

- ☐ I can say what data is and where it comes from.
- ☐ I can ask a clear question and tally every answer once.
- ☐ I can write a sorting rule anybody could follow.
- ☐ I can draw a chart where one square means one.
- ☐ I can read a chart and say one true thing about it.
- ☐ I can check a total and notice when it is wrong.

	confident	almost there	needs more work
asking a clear question	☐	☐	☐
tallying while I asked	☐	☐	☐
writing a rule a partner could use	☐	☐	☐
drawing a chart somebody else could read	☐	☐	☐
saying something true about my data	☐	☐	☐

The class question is the project for this unit.

For teachers

The moment worth waiting for in this unit is a mismatched total. When a pupil's tally does not match the number of people asked, resist fixing it: ask them to find out why. That single conversation teaches more about data integrity than any completed chart, and it is exactly what topic 3.1's worked example is modelling.

Networks and digital communication

The Slate Stone

5 stories · What a network is, Digital communication, Devices that connect, How we use networks, Online safety

A network carries a message, and kindness travels with it.

UNIT 4 STARTS HERE

Joined up, so we can share

Out on the path

This unit is about machines that are joined to each other, and about the messages that travel between them. It is also the unit where we talk about being safe and being kind, because those two things are the same subject.

What Skylark did

Skylark carried a note across the yard and gave it to Pip. Fen said, ***that is not computing.***

It is, though. Skylark is the network. She has two ends, something travels between them, and a person has to read it when it arrives. A real network is that, with wires.

By the end you will

- say what a network is, and what joins one
- follow a message from one end to the other
- name the devices in your room that are joined up
- say what a network lets your class do
- keep private things private, and be kind online

Three sheds, joined with string. Skylark sits on the middle post.

EVERY MESSAGE, IN THREE PARTS

Pip writes it → the network carries it → Moss reads it

A cable is one way through. There are others.

> **Stay safe and kind**
> Everything in this unit happens inside your class, with your teacher there. You will never be asked to send a message to anybody outside the room, and you should never be asked to by anything else either. If you are, stop and tell a trusted adult.

4.1 · WHAT A NETWORK IS

Machines joined so they can share

WAKE YOUR THINKING UP

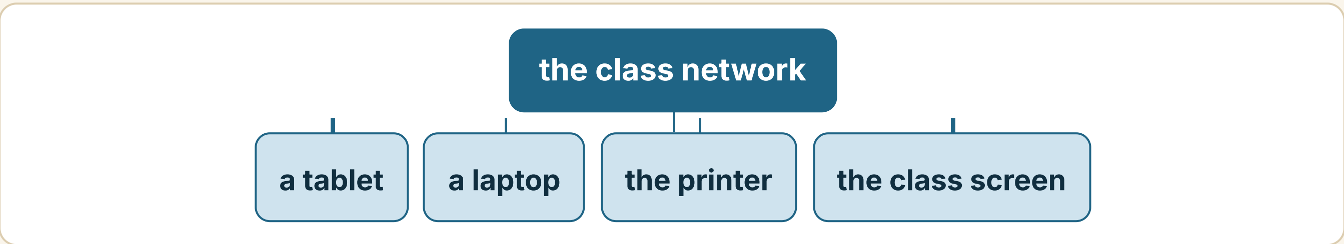
Wake your thinking up. Everybody hold hands in a circle. Squeeze gently, one at a time, round the circle. Now let go of one hand. How far does the squeeze get?

Let us find out

A **network** is machines joined so they can share.

Joined means there is a way through: a wire, or a signal through the air. If there is no way through, there is no network, however close together the machines are.

ONE SHAPE A NETWORK CAN HAVE



Everything joins to one middle point.

AND ANOTHER SHAPE

a tablet — another tablet — a laptop

Here they join to each other, with nothing in the middle. This is still a network.

☞ Watch me try 1

Two tablets sit side by side on a table, switched on. Are they a network?

1. Are they machines? Yes, both of them.
2. Are they close together? Yes, they are touching.
3. Is there a way through, a wire or a signal? No.
4. So they are two machines, not a network.

We found out: being near each other is not being joined, and joined is the only thing that makes a network

How do you know?

How do you know two machines are joined, when you cannot see a wire?

Stay safe and kind

Never plug or unplug a network cable, and never pull one out by the wire. If a cable is loose or damaged, tell a trusted adult.

4.1 · YOUR TURN

Joined, or just near?

1 Ring the answer. Two tablets with no wire and no signal. NO SCREEN

a network | not a network

2 Sort these into the two trays. NO SCREEN

a printer joined by a cable | a book on a shelf | a tablet using the school signal | a pencil case

could be on a network	could not

3 Draw a network of three machines. NO SCREEN

4 Draw the lines that join them. NO SCREEN

5 Build a string network with two chairs and some string. NO SCREEN

6 Cut one string. What can still get through? NO SCREEN

7 Name two things a network lets you share. NO SCREEN

8 Does a network need a machine in the middle? NO SCREEN

9 Draw a network with nothing in the middle. NO SCREEN

10 Which of your two networks keeps working if one part fails? NO SCREEN

11 Name three things joined to your school network. NO SCREEN

12 Why is a network more useful than one machine on its own? NO SCREEN

▣ What you need

string, two chairs, paper, a pencil

☐ No screen needed

String and chairs teach this topic better than any screen. A pupil who has watched a squeeze stop at a let-go hand understands what a broken link is for good.

4.2 · DIGITAL COMMUNICATION

Somebody has to read it

Skylark carries the note. Pip is waiting at the other end with a paw out.

Let us find out

Digital communication is sending a message with a machine. It can be words, a picture, a voice or a video.

Whatever it is made of, it ends the same way: a **person** reads it. That is worth remembering every time you send one.

FOUR STOPS, ONE MESSAGE

Pip's tablet → the class network → the printer → Moss picks up the paper

The message	What it is made of	Who reads it
a note	words	the person you sent it to
a photograph	a picture	everybody you send it to
a voice message	sound	whoever listens
a video	pictures and sound	anybody it is passed on to

☞ Watch me try 2

Fen sends a message and nothing arrives. Where did it stop?

1. Did he write it? Yes, it is on his screen.
2. Did it leave his tablet? The tablet says sent.
3. Did it reach the network? Moss checks: the cable is out.
4. So it stopped at the third stop, before it ever reached Moss.

We found out: a message stops at the first broken link, and finding which one is exactly the same job as debugging

How do you know?

How do you know a message arrived, if nobody replies to it?

Stay safe and kind

Every message you send in this book goes to somebody in your own class, in the same room, with your teacher there. Never send a message to somebody you do not know, and never write anything you would not say out loud to their face.

4.2 · YOUR TURN

Send it, and think who reads it

1 Match each message to what it is made of. NO SCREEN

a note	sound
a photograph	words
a voice message	a picture

2 Write a kind message to somebody in your group. NO SCREEN

3 Read it again. Would you say it out loud to them? NO SCREEN

4 Send it across the room without carrying it. NO SCREEN

5 Draw the stops your message passed through. NO SCREEN

6 Who read it at the far end? NO SCREEN

7 Break one stop. Where does the message get to? NO SCREEN

8 A message arrives with no name on it. What is the problem? NO SCREEN

9 True or not true? NO SCREEN

A message always ends with a person.	**TRUE** / NOT TRUE
A picture can be a message.	**TRUE** / NOT TRUE
Once a message is sent you can always take it back.	**TRUE** / NOT TRUE

10 Name something you should never put in a message. NO SCREEN

11 Why is a message harder to take back than a spoken word? NO SCREEN

12 Write one rule for your class about sending messages. NO SCREEN

What you need

paper, a pencil, string or a message line, a partner

Stay safe and kind

Nothing in your message may be private: no addresses, no telephone numbers, no passwords, and nothing about somebody else that they have not said you may write.

4.3 · DEVICES THAT CONNECT

Every device has a job on the network

Four devices on a bench, each joined to the next.

Let us find out

Lots of different devices join the same network, and each has its own job on it. Some put things in, some put things out, and some only pass things along.

what joins your class network

1 **tablet** you work on it	2 **laptop** the bigger jobs
3 **printer** puts work on paper	4 **class screen** everybody can see it
5 **camera** takes pictures in	6 **the box on the wall** passes everything along

▣ WORDS WE USED

tablet
you work on it

laptop
the bigger jobs

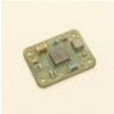
printer
puts work on paper

class screen
everybody sees it

camera
takes pictures in

cable
joins one to another

memory stick
carries files by hand



the box on the wall
passes it all along

☞ Watch me try 3

Is a memory stick part of the network?

1. Does it hold files? Yes.
2. Does it move files from one machine to another? Yes.
3. Is it joined to anything by a wire or a signal? No, you carry it.
4. So it moves files without being on the network at all.

We found out: a memory stick does the same job as a network by being carried, which is why it is useful when there is no network

How do you know?

How do you know a printer is on the network and not just plugged into one laptop?

Stay safe and kind

Devices are carried flat, with two hands, one at a time. Never carry a device by its cable, and never plug anything into a wall socket.

4.3 · YOUR TURN

What joins what?

1 Match each device to its job. NO SCREEN

printer	everybody can see it
camera	joins one to another
class screen	puts work on paper
cable	takes pictures in

2 Sort these into the two trays. NO SCREEN

printer | camera | class screen | keyboard

puts something out	takes something in

3 Count the devices in your room. Write the number. NO SCREEN

I counted

4 Which of them do you think are joined up? NO SCREEN

5 Draw your classroom network. NO SCREEN

6 Label each device on your drawing. NO SCREEN

7 Which one would everybody miss most? Say why. NO SCREEN

8 A memory stick is on the network. True or not true? NO SCREEN

9 Name a device that only passes things along. NO SCREEN

10 Name two devices that could share one printer. NO SCREEN

11 Why is sharing one printer better than having six? NO SCREEN

12 Which device on your drawing puts nothing out at all? NO SCREEN

What you need

this page, a pencil, and your own classroom to look at

Work it out together

In pairs, walk round the room and find every cable you can see without touching any of them. Draw where each one goes. You are mapping a real network.

Stay safe and kind

Look at cables, do not touch them, and never crawl under a desk to follow one. Ask your teacher what is behind anything you cannot see.

4.4 · HOW WE USE NETWORKS

What a network lets your class do

Moss takes the paper. Wren points at the tablet it came from.

Do you remember?

What does joined mean? Say it before you read on.

Let us find out

A network is not interesting on its own. It is interesting because of what it lets a class **do**: save work where everybody can find it, print, and look something up together.

What you do	What the network does
save to the class folder	keeps it where any machine can open it
print your work	carries it to the printer
show work on the big screen	sends it to another device
look something up	fetches it from far away

SAVING TO THE CLASS FOLDER

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT	WHAT IS KEPT
you press save	→	the network carries your file	→	the file appears in the class folder	it is still there tomorrow

☞ Watch me try 4

Wren saves her work at school and opens it the next day. How?

1. She pressed save, which sent her file along the network.
2. The file was kept in the class folder, not on her tablet.
3. The next day she uses a different tablet.
4. The folder is on the network, so any machine can open it.

We found out: her work was not on the tablet at all, which is why a different machine could open it

How do you know?

How do you know your work is saved on the network and not only on the tablet you used?

Stay safe and kind

Save with the name your teacher gives you, never your full name. The class folder is shared, so only put work in it, and never open or change somebody else's file.

4.4 · YOUR TURN

Follow your work

1 Number these from first to last. NO SCREEN

1	2	3	4
it appears in the folder	you press save	the network carries it	you open it tomorrow

2 Where is your work kept: on the tablet or on the network? NO SCREEN

on the tablet | on the network

3 Draw the journey your work takes when you press print. NO SCREEN

4 Name three things your class shares on the network. NO SCREEN

5 The network stops working. Name one thing you could still do. NO SCREEN

6 Name one thing you could not. NO SCREEN

7 Why save with the name your teacher gives you? NO SCREEN

8 Somebody else's file is open. What do you do? NO SCREEN

9 Practise passing one sheet of paper along a row of four. NO SCREEN

10 What happened when the middle person folded their arms? NO SCREEN

11 Which stop is like the middle person in a real network? NO SCREEN

12 Why is a shared folder useful to a whole class? NO SCREEN

What you need

one sheet of paper, a row of four people

Stay safe and kind

Ask before you print, and print once. A shared printer belongs to everybody in the school, and a class that prints twenty copies by accident leaves none for anybody else.

4.5 · ONLINE SAFETY

Private things stay private

Pip turned the tablet over and went to find Moss. That was exactly the right thing to do.

Let us find out

Some things about you are **private**. They are not secrets to be ashamed of, they are simply yours, and they stay yours.

And there is one sentence to remember out of this whole book: if something feels wrong, **stop, and tell a trusted adult**. You will never be in trouble for telling.

SORT THESE IN YOUR HEAD BEFORE YOU READ ON

private, and stays private	fine to share in class

Private	Fine to share in class
your home address	a book you like
your school password	your favourite colour
your telephone number	the football team you support
a photograph of you	what you did in the lesson
how old you are	an answer to a class question

☞ Watch me try 5

A game asks Pip to type her name and age to carry on playing.

1. Is her name private? Her first name in class is fine.
2. Is her age private? Yes, and a game does not need it.
3. Does she have to type it to keep playing? No game is worth that.
4. So she stops, turns the tablet over, and tells a trusted adult.

We found out: she did not have to work out whether the game was safe, because stopping and telling an adult is always the right answer

How do you know?

How do you know whether something is private? What could you ask yourself?

Stay safe and kind

Stop, and tell a trusted adult. Say it out loud with your class. Your trusted adults are your teacher and the grown-ups at home who look after you. If a screen asks you anything about yourself, if a message upsets you, or if somebody you do not know writes to you, you do not deal with it on your own.

4.5 · YOUR TURN

Safe, private and kind

1 Sort these into the two trays. NO SCREEN

my home address | my favourite colour | my school password

a book I like | how old I am | the lesson we had

private	fine to share in class

2 Finish the sentence. If something feels wrong, stop and tell a

NO SCREEN

tell a

3 Write who your trusted adults are.

NO SCREEN

4 A game asks how old you are. Write what you do.

NO SCREEN

5 Somebody you do not know sends you a message. What do you do?

NO SCREEN

6 Rewrite this kindly: your model is rubbish.

NO SCREEN

7 True or not true?

NO SCREEN

A person I only met online is a stranger.	**TRUE** / NOT TRUE
I can share my password with my best friend.	**TRUE** / NOT TRUE
Kind words online are the same as kind words in the room.	**TRUE** / NOT TRUE
Everything on a screen is true.	**TRUE** / NOT TRUE

8 Who is a password for? Write your answer. NO SCREEN

9 You want to post a picture of your partner. What must you do first? NO SCREEN

10 Would you be in trouble for telling an adult? Ring your answer. NO SCREEN

yes no

11 Write one kind sentence and one unkind one. Cross out the unkind one. NO SCREEN

NO SCREEN

12 Why are being safe and being kind the same subject?

What you need

this page, a pencil, and your class to talk with

Work it out together

As a class, make one poster together. On it, write the sentence **stop, and tell a trusted adult**, and the names of the trusted adults in your school. Put it where you can all see it.

Stay safe and kind

Nothing on this page asks you to write anything private down. Notice that: a page that wanted your address would be breaking its own rule.

Staying safe and kind online

for teachers and families

childnet.com

UNIT 4 · WORDS AND CHECKS

Words we used

WORDS WE USED

network
machines joined so they can share

message
something one person sends another

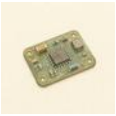
printer
puts work on paper

class screen
everybody can see it

private
not for sharing

trusted adult
a grown-up who looks after you

memory stick
carries files by hand



signal
a way through without a wire

Check what you can do

1. Say what a network is, in your own words.
2. Say what joined means, and name two ways it can happen.
3. Follow a message from one end to the other, naming each stop.
4. Say what a memory stick does that a network also does.
5. Name three private things and three you can share in class.
6. Say the sentence to use if something feels wrong.

Stay safe and kind

The most important thing in this unit is one sentence: stop, and tell a trusted adult. If you remember nothing else from Unit 4, remember that.

UNIT 4 · HOW DID YOU DO?

Look what I can do

- ☐ I can say what a network is and what joins one.
- ☐ I can follow a message and say where it stopped.
- ☐ I can name the devices joined up in my room.
- ☐ I can say what our network lets my class do.
- ☐ I can keep private things private.
- ☐ I can stop, and tell a trusted adult.

	confident	almost there	needs more work
saying what a network is	☐	☐	☐
following a message to the end	☐	☐	☐
naming the devices in the room	☐	☐	☐
sorting private from shareable	☐	☐	☐
knowing what to do if something feels wrong	☐	☐	☐

The message line is the project for this unit.

✸ Skylark's own notes

A message from your school to the other side of the world usually travels most of the way along cables lying on the sea floor. There are hundreds of them, they are about as thick as a hosepipe, and almost everything you look up has been through one.

✎ For teachers

Topic 4.5 is the one to teach slowly and to return to. The aim is not that pupils can recite a list of private things, it is that stopping and telling becomes automatic. Say the sentence with the class, put it on the wall with the names of the actual adults on it, and make sure every pupil knows that telling never gets them into trouble.

Computer systems

The Putty Stone

5 topics · What a system is, Hardware, Software, Working together, Our rules

A computer system is parts and programs working together.

UNIT 5 STARTS HERE

Parts you can hold, and instructions you cannot

Out on the path

This is the last stone on the path. You have written algorithms, built programs, handled data and sent messages. Now you find out what the machine underneath all of it actually is.

What Fen asked

*Moss opened the old machine on his bench. Fen looked inside and said, **where is the program?***

He could see a fan, a board, some wire and a speaker. He could not see the program anywhere, because a program is not a thing you can pick up. That is the whole of this unit in one question.

✓ By the end you will

- say what a computer system is
- name the parts you can hold, and what each one does
- say what software is, and why you cannot hold it
- follow one job all the way through a system
- use the machines in your room well, and leave them ready

Moss and a whole machine, before he opens it.

THE SHAPE OF EVERY JOB A COMPUTER DOES

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT	WHAT IS KEPT
something you do	→	the work inside	→	something you notice	something that is kept

A fan you can hold. A program you cannot.

Stay safe and kind

Never open a machine, and never pull a cable out of a wall socket. Moss is a grown-up badger with the power switched off. Looking inside is something an adult does while you watch.

How the parts of a computer fit together
for teachers, before an unplugged hardware lesson
teachcomputing.org

5.1 · WHAT A SYSTEM IS

Four things happen, every time

WAKE YOUR THINKING UP

Wake your thinking up. Your teacher names a job: taking a photo. The class says what goes in, what happens inside, and what comes out. Three shouts, in order.

Let us find out

A **computer system** is the parts and the programs, working together on one job.

Every job has the same shape. Something goes **in**. Work happens **inside**. Something comes **out**. Sometimes something is **kept**.

PRESSING ONE KEY

WHAT GOES IN		THE WORK INSIDE		WHAT COMES OUT	WHAT IS KEPT
you press a key	→	the machine works out which letter	→	the letter appears on the screen	the letter is kept in your file

The same shape, three times

The job	In	Inside	Out
taking a photo	you press the button	the camera turns light into numbers	a picture on the screen
playing a sound	you tap the sound block	the machine reads the sound file	the speaker moves the air
a robot moving	you press go	it reads the commands one at a time	the wheels turn

☞ **Watch me try 1**

Fen taps Save. What is the input, and what is the output?

1. The input is the tap. That is the thing Fen did.
2. Inside, the machine copies his work into a file.
3. The output is the word Saved appearing on the screen.
4. What is kept is the file, which is still there tomorrow.

We found out: one tap gave an input, some work inside, an output on the screen and one thing kept

How do you know?

Fen says the screen IS the computer. Is Fen right? How do you know?

Stay safe and kind
Two hands on a tablet, and put it down flat before you turn round. Keep every drink away from every keyboard in the room.

5.1 · YOUR TURN

In, inside, out

1 Input or output? Tapping the screen. NO SCREEN

input output

2 Input or output? A picture appears. NO SCREEN

input output

3 Input or output? The speaker plays a pop. NO SCREEN

input output

4 Input or output? You press go on the robot. NO SCREEN

input output

5 Fill in this system. The job is printing your work. NO SCREEN

PRINTING YOUR WORK

WHAT GOES IN	→	THE WORK INSIDE	→	WHAT COMES OUT

6 Now fill one in for a job you choose. NO SCREEN

7 Name one thing your system keeps. NO SCREEN

It keeps

8 Number these from first to last: out, kept, in, inside. NO SCREEN

1	2	3	4
out	kept	in	inside

9 A machine with no output. Would you know it was working? NO SCREEN

10 Name the input when you take a photograph. NO SCREEN

The input is

11 Name the output when you print your work. NO SCREEN

The output is

12 Fen says the screen is the whole computer. Is Fen right? NO SCREEN

▣ What you need

this page, a pencil, a partner

☵ Work it out together

In pairs. One of you is the input, one is the output. Your teacher whispers a job. Act the two ends of it, with no words, and let the class work out what happened inside.

✷ Skylark's own notes

A computer only ever really does one thing: it follows instructions extremely fast. A machine on your teacher's desk can follow more instructions in one second than every person in your school could follow in a whole year.

5.2 · HARDWARE

The parts you can hold

Moss lays the parts out on a cloth, in order, so none of them gets lost.

Let us find out

Hardware is every part of a computer you could pick up. If you can hold it, or drop it, it is hardware.

Some hardware takes things in. Some puts things out. Some does the work inside, and some keeps things for later.

what is in the room

1. **screen** puts pictures out
2. **keyboard** takes letters in
3. **mouse** takes pointing in
4. **speaker** puts sound out
5. **camera** takes pictures in
6. **printer** puts paper out
7. **the board inside** does the work
8. **the memory stick** keeps things

▣ WORDS WE USED

screen
puts pictures out

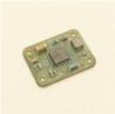
keyboard
takes letters in

mouse
takes pointing in

speaker
puts sound out

camera
takes pictures in

printer
puts paper out



board
does the work inside

memory stick
keeps things

☞ Watch me try 2

Moss holds up the fan. Is it hardware, and what is it for?

1. Can he hold it? Yes, it is in his paw.
2. So it is hardware.
3. It is not taking anything in or putting anything out.
4. It keeps the parts inside cool so they can keep working.

We found out: the fan is hardware, and its job is to stop the working parts getting too hot

How do you know?

Pick up one piece of hardware in your room. How do you know it is hardware and not software?

Stay safe and kind

Look, do not open. The inside of a machine is for an adult, with the power off. If a cable looks damaged, tell a trusted adult and do not touch it.

5.2 · YOUR TURN

Sort the hardware

1 Sort these into the two trays. Write each word once. NO SCREEN

keyboard | screen | camera | speaker | mouse | printer

takes something IN	puts something OUT

2 Which one on this page could do both? Say why. NO SCREEN

3 Match the part to its job. NO SCREEN

keyboard ●	● puts sound out
speaker ●	● does the work
camera ●	● takes letters in
the board inside ●	● takes pictures in

4 Draw one piece of hardware in your room. Label two parts of it. NO SCREEN

5 Name a piece of hardware that keeps things for later. NO SCREEN

6 Wren says a program is hardware. Is Wren right? NO SCREEN

7 Count the pieces of hardware you can see from your seat. NO SCREEN

I can see

8 Which one would you miss most? Say why. NO SCREEN

9 Name a piece of hardware that puts sound out. NO SCREEN

10 Name a piece of hardware that takes pictures in. NO SCREEN

11 Why does a machine need a fan inside it? NO SCREEN

12 Could you use a computer with no output at all? Say why. NO SCREEN

What you need

this page, a pencil, and your own classroom to look at

Skylark's own notes

The first computer mouse was made of wood, in 1964, and it had one button. It was called a mouse because the wire came out of the back of it and looked like a tail.

Stay safe and kind

If you carry hardware, carry one thing at a time and use both paws. A screen that is dropped does not usually work again.

5.3 · SOFTWARE

The instructions you cannot hold

The same laptop. Moss opens one thing, then another, and it does a different job.

Let us find out

Software is the instructions. It is a program. You cannot pick it up, drop it, or put it in a basket.

Here is the surprising part: the **same** hardware does a completely different job when you open different software.

ONE PIECE OF HARDWARE, THREE JOBS

1. one tablet

→

2. open the drawing app: now it is for drawing

→

3. open the block app: now it is for programming

→

4. open the camera: now it takes photographs

Hardware or software?

TWO QUESTIONS, CROSSED

	in your room	not in your room
you can hold it	the screen, the mouse	a machine in another school
you cannot hold it	the drawing app on it	a program nobody has written yet

☞ **Watch me try 3**

Pip deletes the drawing app. Has she broken the tablet?

1. The tablet is hardware. It is still there, in her paws.
2. The app was software: instructions the tablet was following.
3. The instructions are gone, so the tablet cannot draw any more.
4. The tablet itself works perfectly. It just has one less job.

We found out: deleting software does not break hardware: the machine is fine and has one fewer thing it knows how to do

How do you know?

Somebody says a tablet with no software would still be useful. How would you find out if that is true?

Stay safe and kind

Only a teacher installs software. If a page asks you to add or download something, stop and tell a trusted adult. That is their job and never yours.

5.3 · YOUR TURN

Hardware or software?

1 Hardware or software? The keyboard. NO SCREEN

hardware software

2 Hardware or software? The drawing app. NO SCREEN

hardware software

3 Hardware or software? The speaker. NO SCREEN

hardware software

4 Hardware or software? A program you wrote. NO SCREEN

hardware software

5 True or not true? NO SCREEN

Software is instructions.	TRUE / NOT TRUE
You can hold software in your paw.	TRUE / NOT TRUE
The same tablet can do more than one job.	TRUE / NOT TRUE
Deleting an app breaks the tablet.	TRUE / NOT TRUE

6 Name two jobs one tablet in your room can do. NO SCREEN

7 What makes it do a different job? One sentence. NO SCREEN

8 Draw a box for hardware and a box for software. Put four things in each. NO SCREEN

hardware	software

9 Hardware or software? A file with your work in it. NO SCREEN

hardware software

10 The tablet breaks. Is the app still on it? NO SCREEN

11 Name one job your school laptop can do that a robot cannot. NO SCREEN

12 Why can one tablet do many different jobs? NO SCREEN

What you need

this page and a pencil

Path challenge

A program is instructions. Is a recipe software? Say what is the same about them, and what is different.

→ Go a little further

Your algorithm from Unit 1 was instructions too. Was it software? Say why you think so.

5.4 · WORKING TOGETHER

Follow one job all the way through

One job, left to right: a key, the machine, the paper, the file.

Do you remember?

What are the four things that happen in every computer job? Say them in order before you read on.

Let us find out

Hardware on its own does nothing. Software on its own cannot touch anything. A **system** is the two of them together, and you can follow a job right through it.

ONE LETTER, ALL THE WAY THROUGH

Wren presses a key → the board works out which letter → the letter appears on the screen → the file keeps it → the printer puts it on paper

Five stops. Take any one of them away and the job stops there.

☞ Predict, then test

If the printer is switched off, which stop does the job reach?

I think 1	I ran it 2	I compare 3
Write or draw what you think will happen	What really happened?	Was it what you thought? Did it match what happened?

☞ Watch me try 4

Moss taps print and nothing comes out. Where did the job stop?

1. Did the tap work? The screen said printing, so the input was fine.
2. Did the software work? It sent the job, so yes.
3. Did anything reach the printer? Moss checks the cable. It is out.
4. So the job stopped at the fourth stop, before the paper.

We found out: the job travelled as far as the loose cable and stopped there, and the cable was the only thing wrong

How do you know?

How do you know the software was working, even though no paper came out?

Stay safe and kind

Cables are pulled out by the plug, never by the wire. If a machine is hot, or smells odd, or makes a new noise, tell a trusted adult and leave it alone.

5.4 · YOUR TURN

Where did it stop?

Find the bug

THE GOAL

A photograph of Wren's model appears on the class screen.

THE PROGRAM

HOW THE MESSAGE TRAVELS

press the camera button → the camera makes a picture → the picture is kept in a file → the file opens on the class screen

WHAT HAPPENED

The button clicked but no picture was kept.

What is the bug? Change ONE thing, then test it again.

1 Which stop did the job reach? NO SCREEN

It reached stop

2 Name one thing that could be wrong. NO SCREEN

3 How would you check, without changing anything else? NO SCREEN

4 Draw the stops for playing a sound. Use four boxes. NO SCREEN

5 Which stop is the hardware? Which is the software? NO SCREEN

6 Number these from first to last. NO SCREEN

1	2	3	4
the screen shows it	you tap	the file keeps it	the machine works it out

7 Take one stop away from your sound job. What happens? NO SCREEN

8 Why is it useful to know where a job stopped? NO SCREEN

9 Name the LAST stop in the printing job. NO SCREEN

10 Name the FIRST stop in the printing job. NO SCREEN

11 Which stop keeps your work for tomorrow? NO SCREEN

12 A job stops at stop two of five. Has the output happened? NO SCREEN

What you need

this page, a pencil, a partner

Work it out together

In pairs, be a system. One of you is the keyboard, one is the screen. A third person is the board in the middle. Pass one letter along by hand. Now the middle person folds their arms: what reaches the screen?

5.5 · OUR RULES

Leave it ready for the next person

Two of them, one tablet, four paws, walking slowly.

Let us find out

The machines in your room are shared. Somebody used them before you, and somebody will use them after you.

So there are five rules, and all five are about the next person.

OUR FIVE RULES FOR THE MACHINES

1. Two hands, and walk.

→

2. Gentle taps, and no food or drink near it.

→

3. Ask before you open something new.

→

4. Save your work, with your own name on it.

→

5. Put it back where it lives, the right way up.

☞ Watch me try 5

Fen finds a tablet left on a chair, face up, not saved.

1. Whose work is on it? He cannot tell: there is no name on the file.
2. Is it safe there? No. A chair is where somebody sits.
3. What can he do? Save it, so the work is not lost.
4. Then put it back where it lives, and tell the teacher.

We found out: an unsaved tablet on a chair is two rules broken, and saving it first is the kindest thing to do

How do you know?

Why does putting your own name on a file help the next person, and not only you?

Stay safe and kind

Carry one machine at a time, flat, with two hands, and walk. Put it down before you talk to anybody. If you see a machine somewhere it could fall from, tell a trusted adult rather than reaching for it.

No screen needed

Everything in topic 5.5 is done without switching anything on. Practise carrying, putting down, and putting away, with the power off. That is the lesson.

5.5 · YOUR TURN

Good for the next person

1 Sort these into the two trays. NO SCREEN

two hands | run with it | gentle taps | drink beside the keys | save with my name | leave it on a chair

good for the next person	not good for the next person

2 Write the five rules in your own words. NO SCREEN

3 Which rule is hardest to remember? Say why. NO SCREEN

4 Practise carrying a tablet, switched off, across the room. NO SCREEN

5 Practise putting it down and turning round. Did you put it down first? NO SCREEN

I used two hands.	YES / NO
I walked.	YES / NO
I put it down before I turned round.	YES / NO
I put it back where it lives.	YES / NO

6 A friend is about to run with a tablet. Write one kind sentence. NO SCREEN

7 Make a rule card for your computing corner. Draw and write it.

NO SCREEN

8 Why do we save with our own name on the file?

NO SCREEN

9 Which rule stops work getting lost?

NO SCREEN

10 Which rule keeps the machine itself safe?

NO SCREEN

11 You find a tablet on a chair. Write what you do first.

NO SCREEN

12 Why do we ask before opening something new?

NO SCREEN

▣ What you need

a tablet that is switched off, and space to walk

✷ Skylark's own notes

Every computer in every school has a list of rules like these, and so does every office. The rules are almost the same everywhere, because they are not really about computers. They are about sharing.

✎ For teachers

Topic 5.5 is worth teaching with the devices OFF and in pupils' hands. The rules are physical habits rather than facts, and a class that has practised carrying and putting down will break far fewer screens than a class that has only read the list.

UNIT 5 · WORDS AND CHECKS

Words we used

WORDS WE USED

system
parts and programs together

hardware
the parts you can hold

software
the instructions inside

input
what goes in

output
what comes out

storage
what is kept

fan
keeps the parts cool

cable
joins one part to another

☐ Check what you can do

1. Say what a computer system is, in one sentence.
2. Name three pieces of hardware and say what each is for.
3. Say what software is, without using the word program.
4. Name a job one tablet can do two different ways.
5. Follow the job of printing a letter, from the key to the paper.
6. Say one rule that is about the next person, not about you.

! Stay safe and kind

Two rules to carry out of this unit and keep. Never open a machine. If something worries you about a device, stop and tell a trusted adult.

UNIT 5 · HOW DID YOU DO?

Look what I can do

- ☐ I can say what a computer system is.
- ☐ I can name the parts you can hold, and what they do.
- ☐ I can say what software is, and why I cannot hold it.
- ☐ I can follow one job all the way through a system.
- ☐ I can say where a job stopped when it went wrong.
- ☐ I can leave a machine ready for the next person.

	confident	almost there	needs more work
saying what a system is	☐	☐	☐
naming hardware and its jobs	☐	☐	☐
telling hardware from software	☐	☐	☐
following a job all the way through	☐	☐	☐
leaving a machine ready for somebody else	☐	☐	☐

Moss and Fen built their own machine out of cardboard. That is the project for this unit.

Skylark's own notes

You have now crossed all five stones. You can write clear steps, build a program, sort data, send a message and name what is inside the machine doing all of it. That is more computing than most grown-ups can explain.

Five projects along the way

One project for each stone on the path. Each one takes a lesson or two, and each one uses only what you already know by the end of that unit.

The end of the path. All five projects behind you.

After unit	Project	What you make
1	Note carrier	a path that gets a note to the shelf
2	Two-scene story	a program with an event and a repeat
3	Class question	a chart of a real question you asked
4	Message line	a way to get a note across the room
5	Cardboard machine	a machine with inputs and outputs

Every project follows the same five steps, and they are the same five steps you have used all year.

HOW EVERY PROJECT WORKS

1. Plan it →
2. Say what will happen →
3. Build it →
4. Test and fix one thing →
5. Show somebody

Stay safe and kind

Show your plan to your teacher before you build. If your project needs a device, ask first, and never install anything yourself.

The note carrier

AFTER UNIT 1

Note carrier

The goal. Get a folded note from the mat to the bookshelf, using only commands you wrote down before you started.

You may not touch the robot once it is moving, and you may not add a command while it goes. Everything has to be decided first.

Cog with a note on its back, on the way to the shelf.

▣ What you need

a floor robot or a partner as a human robot, arrow cards, a folded note, chalk or tape to mark the start

1 Mark your Start square and your goal. Draw them on the mat.

NO SCREEN

YOUR MAT. DRAW START, THE SHELF, AND ANY OBSTACLES.

2 Write your commands in order.

NO SCREEN

MY COMMANDS

3 Say what will happen. Where will it stop? NO SCREEN

4 Dry-run it with your finger before you run it. NO SCREEN

5 Run it. Did the note arrive? NO SCREEN

6 If not, find the bug. Change ONE thing. NO SCREEN

7 Run it again. Write how many tries it took. NO SCREEN

It took

How do you know?

How do you know your commands were right, and not just lucky?

Path challenge

Do it again in fewer commands, with the note still arriving.

A two-scene story

AFTER UNIT 2

Two-scene story

The goal. Make a story with two places in it. It must start with an event, and something in it must repeat.

Pip shows Wren her first scene.

What you need

a tablet if one is free, or paper and pencils, and a partner

1 Write your story in two sentences. One for each scene. NO SCREEN

2 Draw your two backdrops. Label which comes first. NO SCREEN

scene 1

scene 2

3 Write your script. Start with an event block. NO SCREEN

MY SCRIPT

green flag

4 Where does something repeat? Draw a ring around it. NO SCREEN

5 Say what will happen when you run it. NO SCREEN

6 Build it and run it. Did it match? NO SCREEN

7 Find one thing to improve. Change only that. NO SCREEN

How do you know?

How do you know your repeat saved you some blocks? Count them.

Step-by-step block projects to try next

for teachers and families

projects.raspberrypi.org, Scratch intro

A real class question

AFTER UNIT 3

Class question

The goal. Ask your class a real question, collect every answer once, and show what you found in a chart somebody else can read.

Wren and Moss put their chart on the fence where everybody can see it.

What you need

paper, a pencil, a clipboard if you have one, and your whole class

1 Write your question. It must have a small number of answers. NO SCREEN

2 Write the answers people can choose from. NO SCREEN

3 Ask everybody. Tally as you go. NO SCREEN

MY RESULTS

The answer	Tally	How many

4 Add up each row. Write the totals. NO SCREEN

5 Check your total against the number in your class. NO SCREEN

6 Draw your chart. One square means one. NO SCREEN

7 Write one true sentence about your chart. NO SCREEN

How do you know?

How do you know nobody answered twice?

Stay safe and kind

Ask about things, not about people. A question about favourite fruit is fine. A question about somebody's home, their family or how they look is not, and it is not ours to collect.

A message line

AFTER UNIT 4

Message line

The goal. Build a way to get a written message from one side of the room to the other, without carrying it yourself.

Skylark and Pip send a note along a line strung between two posts.

What you need

string, two chairs, paper clips or pegs, paper, and a partner at the far end

1 Draw your line. Mark both ends and what is in between. NO SCREEN

2 Write the stops your message will pass through. NO SCREEN

3 Write your message. Keep it kind and keep it short. NO SCREEN

4 Send it. Did it arrive? NO SCREEN

5 Whose job was it to read it at the far end? NO SCREEN

6 Break one stop on purpose. Where does the message get to?

NO SCREEN

7 How is your line like a network? Say one way.

NO SCREEN

How do you know?

How do you know a message arrived, if nobody tells you?

Stay safe and kind

Nothing on your message may be private. No addresses, no telephone numbers, no passwords, and nothing about somebody who has not said you may write it.

A cardboard machine

AFTER UNIT 5

Cardboard machine

The goal. Build a machine out of cardboard that has a real input, something happening inside, and a real output. It does not have to switch on. It has to make sense.

Moss and Fen with the machine they made. It has a dial and a lever, and both of them do something.

What you need

a cardboard box, scissors and glue with an adult, split pins, paper for dials and levers, and coloured pencils

1 Decide the job your machine does. Write it in one sentence. NO SCREEN

2 Plan your system. Fill in all four. NO SCREEN

MY MACHINE

WHAT GOES IN	→	THE WORK INSIDE	→	WHAT COMES OUT

3 Draw your machine from the front. Label the input and the output. NO SCREEN

4 Build it. Ask an adult for the cutting. NO SCREEN

5 Show a partner. Can they find the input without being told? NO SCREEN

6 Fix one thing they could not work out. NO SCREEN

7 Which parts of your machine are hardware? NO SCREEN

8 Where is the software in your machine? Say what it would be. NO SCREEN

How do you know?

How do you know your output really is an output?

Stay safe and kind

An adult does all the cutting. Split pins are sharp: push them through with the point away from your paw.

Our own computing project

THE END OF THE PATH

Choose your own

This is the big one, and you choose it. It uses whatever you like from the whole year, and it is yours: your goal, your plan, your bugs and your fixes.

Letter	Project	It uses
A	A delivery robot	unit 1 and unit 2
B	A helper program	unit 2
C	A class survey and a chart	unit 3
D	A message system for the room	unit 4
E	A machine you build	unit 5
F	Your own idea	you decide, and tell your teacher

1 Which letter did you choose? NO SCREEN

I chose

2 Write your title. NO SCREEN

3 Write your goal in ONE sentence. A weak goal is: make something good. NO SCREEN

4 How will you know it worked? Write your success check. NO SCREEN

5 Write your six steps, from first to last. NO SCREEN

MY PLAN

1

→

2

→

3

→

4

→

5

→

6

▣ What you need

your plan, whatever your project needs, and a partner to test with

A goal you can point at. That is the whole trick.

☛ **Watch me try 1**

Fen's goal was: make a good robot thing. Is that a goal?

1. Could somebody else tell when it was finished? No.
2. Could Fen tell? Not really.
3. So it is a wish, not a goal.
4. Better: get a note from the mat to the shelf without touching Cog.

We found out: a goal has to be something you can point at and say yes, that happened, or it cannot be tested

Build it, test it, show it

6 Build the first part only. Test that before you go on. NO SCREEN

7 Something will go wrong. Write the bug. NO SCREEN

8 Write the fix. Change ONE thing. NO SCREEN

9 Write the cause. Why did the bug happen? NO SCREEN

10 Test it again. Did your success check pass? NO SCREEN

11 Show your project. Say these three things. NO SCREEN

WHAT TO SAY WHEN YOU SHOW IT

1 My goal was... → 2 The step I am proudest of is... → 3 The bug I fixed was...

12 Write one thing you would do differently next time. NO SCREEN

How do you know?

Somebody says your project worked. How do you know they are right? Point at your success check.

Show it to somebody who has not seen it before.

What you need

your project, your plan, and somebody who has not seen it before

Work it out together

Show your project to a partner from a different group. They get to ask you two questions, and one of them has to be **how did you fix that?**

	confident	almost there	needs more work
planning before building	☐	☐	☐
saying what would happen first	☐	☐	☐
finding my own bug	☐	☐	☐
changing one thing at a time	☐	☐	☐
showing my work to somebody else	☐	☐	☐

✷ Skylark's own notes

Real programmers work exactly like this, and they show each other their work before it is finished on purpose. Somebody who has not seen it before finds things you cannot see any more, because you already know what it is meant to do.

✎ For teachers

The success check in task 4 is the part worth insisting on. A pupil with a testable check can evaluate their own project honestly, and a pupil without one can only ask whether the teacher liked it. Refuse goals that cannot be pointed at.

What you can do so far

This covers Unit 1 and the start of Unit 2: algorithms, clear steps, bugs, scripts, motion and events.

NAME ______________________ DATE __________

HOW TO WORK

1. Listen to your teacher.
2. Say the answer out loud first.
3. Then write, ring or draw it.
4. Ask for help if you need it. That is allowed.

1 What is an algorithm? Ring the best answer. NO SCREEN

a snack | clear steps for a job | a kind of robot

2 Make this clear: **go over there.** NO SCREEN

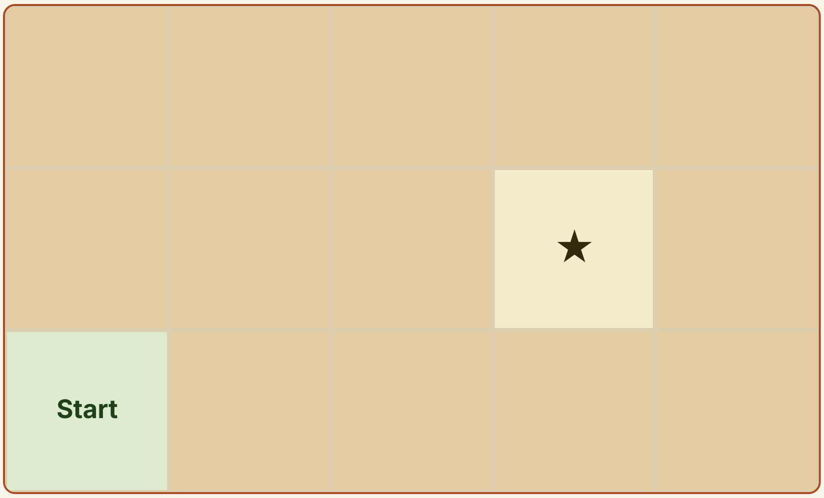
3 Number these from first to last. NO SCREEN

1	2	3	4
close the lid	put the book in	open the box	pick up the book

4 Ring the one that is NOT a kind of bug. NO SCREEN

a missing step | the wrong order | a tidy desk

5 Cog should stop on the star. Draw where it really stops. NO SCREEN



↑ → → → ■

6 What is the bug? Write it in a few words. NO SCREEN

7 How many things do you change when you fix a bug? NO SCREEN

one | four | all of them

Blocks and events

8 This script will not run. What is missing? NO SCREEN

→ move right 2
say hello

9 Ring the block that STARTS a script. NO SCREEN

move right | green flag | say hello

10 What does the number on a move block tell you? NO SCREEN

11 Draw a ring around the two blocks that are NOT joined. NO SCREEN

green flag
→ move right 3

12 True or not true?

NO SCREEN

A robot knows what you meant.	**TRUE** / NOT TRUE
A bug is a mistake in the steps.	**TRUE** / NOT TRUE
An event goes at the top of a script.	**TRUE** / NOT TRUE
Blocks work even when they do not touch.	**TRUE** / NOT TRUE

13 Write one thing you are proud of learning this term.

NO SCREEN

- [ ] I can write clear steps.
- [ ] I can find a bug and say what kind it is.
- [ ] I can start a script with an event.

For teachers

Formative only, and worth saying so to the class before they start. Question guide: 1 to 3 algorithms and order, 4 to 7 debugging, 8 to 11 scripts and events, 12 misconceptions, 13 pupil voice. A pupil who gets 12 right has understood the term's most important idea, whatever else they missed.

TERM 2 CHECK

Programs and data

> **This covers** the rest of Unit 2 and all of Unit 3: loops, sensing, variables, lists, sorting, tallying and charts.

NAME ____________________ DATE __________

HOW TO WORK

1. Listen to your teacher.
2. Say the answer out loud first.
3. Then write, ring or draw it.
4. Ask for help if you need it. That is allowed.

1 How many claps does this make? NO SCREEN

```
↻ repeat [3] times
    ♪ clap
```

2 Write the unit that repeats. NO SCREEN

The unit is

↑ → | ↑ → | ↑ →

3 Why is a loop useful? One sentence. NO SCREEN

4 A variable box holds 4. You add 1. What is in the box now? NO SCREEN

score

1	2
4	

5 Where is the old number now? NO SCREEN

6 This list is in order. What is in position 3? NO SCREEN

my list

1	2	3	4
red	blue	green	yellow

7 Ring the sorting rule that is clear enough to use. NO SCREEN

things I like | has wheels or no wheels | nice ones

Reading what you collected

8 Fill in the totals. NO SCREEN

FAVOURITE CLASS PET

What we found	Tally	How many
cat	/	
dog	////	
fish	//	

9 Which has the most? NO SCREEN

The most is

10 Which has the least? NO SCREEN

The least is

11 Draw the chart. One square means one. NO SCREEN

12 Write one true sentence about your chart. NO SCREEN

13 Twelve people voted, but your total is fourteen. What went wrong? NO SCREEN

14 Why does one person get one vote? NO SCREEN

- [ ] I can use a loop instead of repeating myself.
- [ ] I can tally a real question and total it.
- [ ] I can read a chart and say something true about it.

For teachers

Question guide: 1 to 3 loops and units, 4 to 6 variables and lists, 7 sorting rules, 8 to 12 data handling, 13 to 14 the integrity of a count. Question 13 is the most revealing on the page: a pupil who says somebody voted twice has understood what data IS.

Networks, systems and staying safe

This covers Unit 4 and Unit 5: networks, messages, hardware, software, systems and our rules.

NAME ______________________ DATE ____________

HOW TO WORK

1. Listen to your teacher.
2. Say the answer out loud first.
3. Then write, ring or draw it.
4. Ask for help if you need it. That is allowed.

1 What is a network? Ring the best answer. NO SCREEN

machines joined so they can share

a kind of chart

a program with a loop

2 Ring the two things that could join a network. NO SCREEN a printer | a sandwich | a tablet

3 Hardware or software? Write H or S beside each. NO SCREEN

the keyboard	
the drawing app	
the speaker	
a program	

4 Name one thing you cannot hold. NO SCREEN

5 Fill in this system. The job is printing. NO SCREEN

PRINTING

WHAT GOES IN → THE WORK INSIDE → WHAT COMES OUT

6 The paper never came out. Where did the job stop? NO SCREEN

Private, kind, and ready for the next person

7 Sort these into the two trays. NO SCREEN

my school password | my favourite colour | my home address | the name of a book I like

private	fine to share in class

8 Finish the sentence. If something feels wrong, stop and tell a NO SCREEN

tell a

9 Somebody online asks how old you are. What do you do? NO SCREEN

10 Rewrite this kindly: your picture is rubbish. NO SCREEN

11 True or not true? NO SCREEN

Everything on a screen is true.	TRUE / NOT TRUE
A person I only met online is a stranger.	TRUE / NOT TRUE
I can share a password with my best friend.	TRUE / NOT TRUE
Kind words online are the same as kind words in the room.	TRUE / NOT TRUE

12 Write one rule that is about the next person, not about you. NO SCREEN

13 Write one thing you can do now that you could not do in September. NO SCREEN

- [ ] I can say what a network is and what joins it.
- [ ] I can tell hardware from software.
- [ ] I can keep private things private.
- [ ] I can say who my trusted adult is.

	confident	almost there	needs more work
naming the parts of a system	☐	☐	☐
following a job all the way through	☐	☐	☐
keeping private things private	☐	☐	☐
using kind words	☐	☐	☐
leaving a machine ready for somebody else	☐	☐	☐

✎ For teachers

Question guide: 1 to 2 networks, 3 to 6 systems, 7 to 11 e-safety and kindness, 12 to 13 responsibility and pupil voice. Questions 7 to 11 are the ones to follow up individually. A pupil who ticks that a password may be shared with a best friend has told you something more useful than a mark, and it is worth five minutes the same week.

Everything you can do now

Packed up at the end of the year.

Stone	Unit	What you can do
terracotta	1 Computational thinking	write clear steps, predict, find a bug, name its cause
ochre	2 Coding and programming	join blocks in order, use events, loops, variables and lists
sage	3 Managing data	ask a question, tally, sort by a clear rule, draw and read a chart
slate	4 Networks and communication	say what a network is, send a message, keep private things private
putty	5 Computer systems	name hardware and software, follow a job through a system

- [ ] I can write instructions that are clear and exact.
- [ ] I can say what will happen before I test it.
- [ ] I can find a bug, fix one thing, and say why it happened.
- [ ] I can build a program with an event and a repeat.
- [ ] I can collect real data and show it so others can read it.
- [ ] I can send a message and think about who reads it.
- [ ] I can name what is inside a computer, and what is not.
- [ ] I can look after a shared machine.
- [ ] I can stop and tell a trusted adult.

✷ Skylark's own notes

Every one of those skills is used by grown-ups who build the software on the machines in your house. They do the same five things you did this year: plan, predict, build, test and fix. They just do them for longer.

Every word in this book

Say each word out loud. Put your finger under it. If you meet a word on a page and cannot remember it, it is here, and looking it up is what a glossary is for.

algorithm
clear steps for a job

app
a program you open to do one kind of job

arrow cards
arrow cards let you plan with no screen

backdrop
the picture behind a character

block
one instruction you can join to another

board
the part inside that does the work

bug
a mistake in the steps

cable
a wire that joins one part to another

camera
the part that takes pictures in

cause
the reason a bug happened

character
who moves on the stage

chart
a picture of data

class screen
a big screen everybody can see

clear
easy to follow, with only one meaning

command
one instruction for a robot

condition
a question with a yes or no answer

data
information we collected on purpose

debug
find a bug and fix it

device
a machine you can use, like a tablet

dry run
saying the steps before you test them

event
something that starts a program

fan
the part that keeps the inside cool

file
saved work you can open again later

floor mat
the squares a floor robot drives on

floor robot
a robot that follows a path on the floor

goal
what success looks like

hardware
the parts of a computer you can hold

input
what goes in, like a tap or a key

interface
the parts of a program you look at and touch

keyboard
the keys you press to put letters in

kind
friendly and respectful, online and in the room

laptop
a computer that folds shut, for the bigger jobs

least
the smallest group

list
boxes kept in order, so position matters

loop
doing the same steps again on purpose

mat
the squares a floor robot drives on

memory stick
a small part that keeps files

message
something one person sends another

most
the biggest group

mouse
it takes pointing in

network
machines joined so they can share

obstacle
something in the way

pencil
still the fastest tool for planning

personal information
private facts about you

plan
the steps you write before you build

precise
clear and exact: how many, which way, when to stop

predict
say what will happen before you test it

printer
a part that puts your work on paper

order
first, next, last

output
what comes out, like a picture or a sound

paper plan
paper is where a program starts

password
a secret word that keeps your work yours

path
the way a robot goes

pattern
something that repeats

private
not for sharing

program
steps a computer or robot can run

project
a longer making job

purpose
the reason you are making something

repeat
do it again

robot
a machine that follows the steps it is given

rule
something we all do, to keep each other safe

safe
careful, so people and machines are all right

save
keep your work so you can open it later

screen
the part that shows you pictures and words

script
blocks joined in order, ready to run

sensing
when a program can tell something has happened

sequence
steps in order

share
let somebody else see it

signal
a way through the air, with no wire

software
the instructions inside a computer

sort
put into groups using a clear rule

sound
something you can hear coming out

speaker
the part that puts sound out

stage
where you watch a program happen

storage
what is kept for later

system
the parts and the programs, working on one job

table
information in rows and columns

tablet
a flat computer you work on by touching it

tally
one mark for each thing you count

test
try it and watch what really happens

the box on the wall
it passes messages along the network

tool
anything that helps you do a job

total
how many altogether

trusted adult
a grown-up whose job is to look after you

variable
a labelled box that holds one thing at a time

How this book was built

✎ For teachers

The curriculum contract. This book teaches all 31 topics of the Edu360 Computing Year 2 topic list, in five units, and the build refuses to produce a PDF until every one of those topics is claimed by a page. Nothing has been quietly dropped or merged.

The structure, and why it differs from the manuscript. The Year 2 student manuscript is arranged as thirteen weekly chapters. This book is arranged by strand, because that is how the Edu360 list is written and how progress in it is reported. Every chapter of the manuscript is here: chapter 1 sits in Unit 5, chapters 2 to 5 in Unit 1, chapters 6 to 9 in Unit 2, chapter 10 in Unit 3, chapter 11 across Units 4 and 5, chapter 12 as the final project and chapter 13 as the skills summary. The Initiation chapter is front matter, where a first-lesson introduction belongs.

Forty per cent of the practice needs no device, and that is measured. A Year 2 class typically shares six tablets between twenty-four pupils. Every task that works with cards, chalk, paper or a partner is marked *no screen*, and the build counts them per unit against a floor. If the trolley is booked, the lesson still happens.

Prediction before running is not optional. Every unit asks pupils to say what will happen, run it, and then compare the two. The comparison is the part that teaches, and a topic that predicts without comparing fails the build. Pupils who are only asked to run a program learn to guess.

For teachers, continued

Misconceptions this book tackles head on. That a computer thinks or decides for itself; that a robot knows what you meant; that a machine is alive or can feel; that a bug means the pupil is bad at computing; that the computer made a mistake by itself; that algorithms are only a computer thing; that anything on a screen is true; that a person met online is a friend; that deleting something removes it everywhere; and that a password may be shared with a best friend. Each one is put in a character's mouth or inside a question so pupils refute it, and the book never asserts one in its own voice.

E-safety is taught positively and it is gated. No task in this book asks a pupil to enter a real name, address, age, school, photograph or password, to create an account, to install anything, or to contact anybody outside the class. Units 4 and 5 carry a safety panel on every topic. The phrase pupils are given is the one that works: stop, and tell a trusted adult.

The QR codes. There are twelve, each with a stated educational purpose and each printed with its address in readable words as well, so a page works with no device. The source manuscript used a YouTube search URL per chapter; those were removed. A search page is not a destination, its results change without notice, and sending a family to unvetted results contradicts the e-safety unit in this same book.

Assessment. The three term checks are formative. They exist so a teacher can see what has landed, and no item on them repeats a practice item from the units. The self-assessment wording throughout is confident, almost there, needs more work: deliberately not a traffic light, which tells a child a colour rather than a next step.

Cambridge Early Years

for teachers

cambridgeinternational.org, Cambridge Early Years

Where to go next

✎ **For teachers**

Running the units in a different order. Units 1 and 2 are sequential: Unit 2 assumes pupils can already write and debug a sequence. Units 3, 4 and 5 are independent of each other and may be taught in any order after Unit 1. Unit 5 works well early in the year if the class is curious about the machines themselves.

If you have no floor robot. Every robot activity in this book works with a human robot: one pupil gives commands, another follows them exactly and nothing else. The learning is identical and the mistakes are funnier, which helps. The squared mats print at a size that works with chalk on the playground.

If you have no tablets at all. Unit 2 can be taught entirely on paper with cut-out block cards. Pupils lay out a script on the desk, and a partner acts as the machine and follows it literally. This is how the unit was trialled, and pupils who plan on paper first write better programs when they do reach a screen.

The three destinations worth knowing. The QR codes on the front matter, in Unit 1 and in the back matter open, in order: activities that need no computer, primary lesson plans for algorithms and debugging, and block projects for pupils who finish early. Between them they cover most of what a teacher needs beyond these pages.

What comes after this book. Pupils leaving Year 2 with this material can write and debug a sequence, use an event and a loop, handle a variable, collect and display data, and talk about a system. That is the ground a Year 3 course builds on, and the habit that matters most is the one in the middle: predict, run, compare.

FOR TEACHERS · CONTINUED

Puzzles that use the blocks you know
blockly.games

If something online worries you
ceop.police.uk, safety centre

The person who wrote the first program
Ada Lovelace, on Wikipedia

If a pupil struggles with	Go back to	And try
clear instructions	topic 1.1	acting the steps out with a partner following literally
finding a bug	topic 1.4	the five questions, out loud, in order, every time
loops	topic 2.6	clapping the unit before drawing it
variables	topic 2.8	one real box and one counter, physically swapped
reading a chart	topic 3.5	one true sentence, said aloud, before writing anything
hardware and software	topic 5.3	asking only: could I hold it?

PRIME BOOKS

Computing & Robotics

Year 2 · Cambridge Early Years · Student Manual

A robot does exactly what you say. Not what you meant.

That one idea runs through this whole book. Cross five stepping stones with Fen, Pip and Cog: write instructions that are clear and exact, build a program out of blocks, collect real data from your own class, send a message across the room, and find out what is actually inside the machine doing all of it.

Nearly two thirds of the practice in this book needs no screen at all, so the lesson happens whether the tablets are free or not.

INSIDE THIS BOOK

- All 31 Computing Year 2 topics, in five units
- Predict, run and compare on every unit
- Real bug hunts, with the fix never printed
- Five projects along the way, and one big one at the end
- Three term checks, and a glossary of every word
- Twelve QR codes, each printed in words as well

Prime Books · Computing & Robotics

Ages 6–7 · Lower Primary

primeschool.pt

9 789893 374207